Rank	Country	ISO	$\bar{\chi}$	W	CV(χ)	Regime
1	Germany	DEU	0.55129	0.056714	0.7572	B-M
2	Japan	JPN	0.449327	0.127716	0.5011	C-M
3	Canada	CAN	0.412878	0.251945	0.5015	C-M
4	Bermuda	BMU	0.381129	1.761957	0.3873	C-U
5	Taiwan	TWN	0.458769	0.164548	0.6976	C-M
6	France	FRA	0.433586	0.198433	0.6622	C-M
7	Brazil	BRA	0.467487	0.213593	0.7962	C-M
8	Tajikistan	TJK	0.302466	0.07537	0.1652	C-U
9	Argentina	ARG	0.482249	0.117873	0.8956	C-M
10	Pakistan	PAK	0.423321	0.060141	0.6976	C-M
11	Netherlands	NLD	0.424725	0.109249	0.7636	C-M
12	Russia	RUS	0.409761	0.022095	0.7126	C-M
13	Slovenia	SVN	0.398913	0.014341	0.6733	C-M
14	China	CHN	0.350969	0.369739	0.477	C-U
15	Czechia	CZE	0.420489	0.020877	0.7861	C-M
16	Bulgaria	BGR	0.418988	0.147644	0.8214	C-M
17	Mexico	MEX	0.36743	0.342179	0.6226	C-M
18	South Korea	KOR	0.350969	0.188146	0.5548	C-M
19	Slovakia	SVK	0.474198	0.078263	1.1443	C-B
20	India	IND	0.378048	0.426668	0.7538	C-M
21	Belarus	BLR	0.373137	0.075613	0.7588	C-M
22	Iran	IRN	0.378956	0.138855	0.8102	C-M
23	Lebanon	LBN	0.280742	0.078117	0.3631	C-U
24	Romania	ROU	0.34095	0.051869	0.6604	C-M
25	Nepal	NPL	0.289348	0.038146	0.4098	C-U
26	Saint Vincent and the Gren:	VCT	0.337959	1.080461	0.6704	C-M
27	Ukraine	UKR	0.392424	0.060886	0.9414	C-M
28	Kazakhstan	KAZ	0.303	0.077206	0.5053	C-M
29	United States	USA	0.332278	0.097675	0.6516	C-M
30	Cape Verde	CPV	0.285568	0.045648	0.4403	C-U
31	United Kingdom	GBR	0.421085	0.033011	1.1238	C-B
32	Haiti	HTI	0.315485	0	0.6216	C-M
33	Finland	FIN	0.414148	0.304646	1.1434	C-B
34	Spain	ESP	0.337017	0.104321	0.7567	C-M
35	Belgium	BEL	0.401536	0.121199	1.1114	C-B
36	Hungary	HUN	0.404276	0.07583	1.1419	C-B
37	Democratic Republic of Cor	COD	0.253498	0.028668	0.3485	C-U
38	Guyana	GUY	0.239328	0.09665	0.2814	C-U
39	Malta	MLT	0.293648	0.016632	0.5968	C-M
40	Mauritania	MRT	0.253454	0.042127	0.3947	C-U
41	Azerbaijan	AZE	0.276258	0.039319	0.5228	C-M
42	Senegal	SEN	0.29293	0.043515	0.6187	C-M
43	Puerto Rico	PRI	0.336281	0.028483	0.8681	C-M

44	Central African Republic	CAF	0.421925	0.116404	1.3448	C-B
45	Dominica	DMA	0.29712	1.796673	0.6622	C-M
46	Oman	OMN	0.231744	0.109005	0.2961	C-U
47	Zambia	ZMB	0.234839	0.062843	0.3257	C-U
48	Botswana	BWA	0.252584	0.066964	0.4385	C-U
49	Mozambique	MOZ	0.245145	0.043976	0.4007	C-U
50	Cote d'Ivoire	CIV	0.235096	0.19764	0.3495	C-U
51	Barbados	BRB	0.234103	0.158872	0.3464	C-U
52	Mauritius	MUS	0.277644	0.144253	0.5962	C-M
53	Algeria	DZA	0.273504	0.141916	0.5757	C-M
54	Indonesia	IDN	0.3337	0.095755	0.9307	C-M
55	Mali	MLI	0.231255	0.056922	0.3393	C-U
56	Burkina Faso	BFA	0.225844	0.269131	0.3289	C-U
57	Serbia	SRB	0.298783	0.159838	0.7639	C-M
58	Namibia	NAM	0.254446	0.015479	0.5135	C-M
59	Panama	PAN	0.32075	0.08699	0.9156	C-M
60	Afghanistan	AFG	0.232313	0.235548	0.393	C-U
61	Norway	NOR	0.275003	0.117632	0.6505	C-M
62	Luxembourg	LUX	0.284868	0.065598	0.7197	C-M
63	Greenland	GRL	0.26786	2.461628	0.6245	C-M
64	Hong Kong	HKG	0.239356	0.222669	0.4591	C-U
65	Sierra Leone	SLE	0.268523	0.126566	0.6386	C-M
66	North Macedonia	MKD	0.337109	0.068225	1.0644	C-B
67	Sao Tome and Principe	STP	0.300591	0.071402	0.8414	C-M
68	Eswatini	SWZ	0.233361	0.101111	0.4308	C-U
69	Moldova	MDA	0.254047	0.060496	0.5575	C-M
70	Sweden	SWE	0.351993	0.072833	1.1577	C-B
71	Papua New Guinea	PNG	0.24607	0.218278	0.5127	C-M
72	Belize	BLZ	0.249634	0.027065	0.5358	C-M
73	Syria	SYR	0.240953	0.204306	0.4828	C-U
74	Guatemala	GTM	0.288322	0.092418	0.7768	C-M
75	Tunisia	TUN	0.242968	0.099981	0.5114	C-M
76	Peru	PER	0.310687	0.026215	0.9347	C-M
77	Ecuador	ECU	0.24266	0.035768	0.5152	C-M
78	El Salvador	SLV	0.234325	0.029789	0.4673	C-U
79	Trinidad and Tobago	TTO	0.230685	0.135208	0.4485	C-U
80	Malawi	MWI	0.255737	0.039511	0.6061	C-M
81	Albania	ALB	0.287248	0.052028	0.8064	C-M
82	Nicaragua	NIC	0.253153	0.028639	0.5985	C-M
83	Jamaica	JAM	0.261248	0.020486	0.65	C-M
84	Georgia	GEO	0.224895	0.059649	0.422	C-U
85	Fiji	FJI	0.246454	0.05352	0.5596	C-M
86	Macao	MAC	0.242326	0.028816	0.5371	C-M
87	Costa Rica	CRI	0.225339	0.093206	0.434	C-U
88	New Caledonia	NCL	0.225071	0.107795	0.4352	C-U

89	Niger	NER	0.215101	0.210778	0.3723	C-U
90	Zimbabwe	ZWE	0.224702	0.080726	0.4366	C-U
91	Brunei	BRN	0.202471	0.285093	0.2969	C-U
92	Morocco	MAR	0.292601	0.056055	0.8829	C-M
93	Bolivia	BOL	0.266015	0.064974	0.7133	C-M
94	Faroe Islands	FRO	0.21784	0.105941	0.4058	C-U
95	Latvia	LVA	0.238772	0.092754	0.5405	C-M
96	South Africa	ZAF	0.297025	0.067304	0.9172	C-M
97	Myanmar	MMR	0.253853	0.089198	0.6439	C-M
98	Mongolia	MNG	0.233982	0.022926	0.5177	C-M
99	Equatorial Guinea	GNQ	0.230616	0.992699	0.4978	C-U
100	Nigeria	NGA	0.219846	0.128043	0.4382	C-U
101	Suriname	SUR	0.253864	0.143873	0.6611	C-M
102	Venezuela	VEN	0.242952	0.247031	0.6066	C-M
103	New Zealand	NZL	0.282624	0.309182	0.8792	C-M
104	Angola	AGO	0.219026	0.155782	0.4582	C-U
105	Dominican Republic	DOM	0.282267	0.01642	0.9286	C-M
106	North Korea	PRK	0.219566	0.260367	0.5062	C-M
107	Uzbekistan	UZB	0.222779	0.243292	0.5522	C-M
108	Jordan	JOR	0.229132	0.099533	0.5974	C-M
109	Uganda	UGA	0.232107	0.092299	0.6302	C-M
110	Honduras	HND	0.237952	0.038519	0.6745	C-M
111	Madagascar	MDG	0.235056	0.210454	0.6657	C-M
112	Denmark	DNK	0.246457	0.032011	0.753	C-M
113	Togo	TGO	0.219935	0.146231	0.5638	C-M
114	Estonia	EST	0.233536	0.039105	0.6664	C-M
115	Rwanda	RWA	0.224136	0.06438	0.6026	C-M
116	Poland	POL	0.252943	0.026823	0.8108	C-M
117	Iraq	IRQ	0.225773	0.249844	0.6223	C-M
118	Kuwait	KWT	0.194784	0.109839	0.4032	C-U
119	Colombia	COL	0.256606	0.088706	0.8532	C-M
120	Thailand	THA	0.271213	0.13428	0.96	C-M
121	Ireland	IRL	0.269929	0.266615	0.9584	C-M
122	Cameroon	CMR	0.232225	0.020643	0.69	C-M
123	Lithuania	LTU	0.260297	0.067938	0.8938	C-M
124	Greece	GRC	0.266349	0.056941	0.9493	C-M
125	Laos	LAO	0.262722	0.044063	0.9282	C-M
126	Vietnam	VNM	0.252593	0.132046	0.8535	C-M
127	Tanzania	TZA	0.217559	0.035481	0.6027	C-M
128	Bangladesh	BGD	0.229598	0.099434	0.6982	C-M
129	Ethiopia	ETH	0.214851	0.060379	0.5999	C-M
130	Kyrgyzstan	KGZ	0.211789	0.238691	0.5835	C-M
131	French Polynesia	PYF	0.193235	0.060864	0.4492	C-U
132	Congo	COG	0.223632	0.01319	0.6915	C-M
133	Philippines	PHL	0.277688	0.085178	1.1295	C-B

134	Ghana	GHA	0.213213	0.075718	0.6427	C-M
135	Cambodia	KHM	0.210231	0.017223	0.6249	C-M
136	Switzerland	CHE	0.285346	0.056275	1.2107	C-B
137	Sudan	SDN	0.203096	0.324653	0.5887	C-M
138	Kenya	KEN	0.20494	0.056865	0.6122	C-M
139	Israel	ISR	0.246562	0.010085	0.9562	C-M
140	Guinea	GIN	0.190275	0.326504	0.5408	C-M
141	Uruguay	URY	0.211497	0.008645	0.723	C-M
142	Cuba	CUB	0.227254	0.174494	0.8736	C-M
143	Bahrain	BHR	0.237682	0.07634	0.9611	C-M
144	Austria	AUT	0.220298	0.105385	0.8744	C-M
145	Armenia	ARM	0.196215	0.168685	0.6718	C-M
146	Seychelles	SYC	0.253623	0.020457	1.1631	C-B
147	Malaysia	MYS	0.214922	0.045393	0.8555	C-M
148	Croatia	HRV	0.225683	0.096766	0.9532	C-M
149	Turkey	TUR	0.246363	0.135003	1.163	C-B
150	Samoa	WSM	0.206273	0.152973	0.831	C-M
151	Chile	CHL	0.268396	0.081541	1.4161	C-B
152	Libya	LBY	0.178013	0.453314	0.6019	C-M
153	Italy	ITA	0.234979	0.136984	1.1352	C-B
154	Iceland	ISL	0.21595	0.092568	0.983	C-M
155	Burundi	BDI	0.223981	0.011015	1.2199	C-B
156	Cyprus	CYP	0.215761	0.021275	1.1813	C-B
157	Vanuatu	VUT	0.256538	0.036726	1.6695	C-B
158	Benin	BEN	0.191792	0.027136	1.1603	C-B
159	Aruba	ABW	0.174277	0.033416	1.0232	C-B
160	Portugal	PRT	0.205342	0.069815	1.4134	C-B
161	Egypt	EGY	0.159076	0.072221	0.8747	C-M
162	Paraguay	PRY	0.176806	0.032034	1.1398	C-B
163	Yemen	YEM	0.163512	0.029622	1.0105	C-B
164	Antigua and Barbuda	ATG	0.234267	0.070655	1.9048	C-B
165	Gabon	GAB	0.124413	0.891972	0.5434	A-M
166	Qatar	QAT	0.170253	0.045775	1.1804	C-B
167	Saint Kitts and Nevis	KNA	0.235243	0.069706	2.1307	C-B
168	Chad	TCD	0.164619	0.03445	1.3198	C-B
169	Palestine	PSE	0.169832	0.06136	1.4078	C-B
170	Maldives	MDV	0.161078	0.065729	1.309	C-B
171	Guam	GUM	0.16175	0.065749	1.394	C-B
172	Falkland Islands	FLK	0.313588	0.059925	3.6528	C-B
173	Gibraltar	GIB	0.312976	0.060108	3.7975	C-B
174	United States Virgin Islands	VIR	0.148645	0.070495	1.309	A-B
175	American Samoa	ASM	0.285665	0.068799	3.8165	C-B
176	Eritrea	ERI	0.155044	0.064735	1.7819	C-B
177	Somalia	SOM	0.164009	0.033905	2.0236	C-B
178	Liberia	LBR	0.144546	0.060521	1.941	A-B

179	Cayman Islands	CYM	0.141033	0.071755	2.0331	A-B
180	Cook Islands	COK	0.144866	0.056188	2.188	A-B
181	Saint Lucia	LCA	0.131132	0.071811	2.3061	A-B
182	Bahamas	BHS	0.140471	0.077455	2.7822	A-B
183	Tonga	TON	0.137624	0.063734	2.8299	A-B
184	Turkmenistan	TKM	0.129434	0.100446	2.7881	A-B
185	Australia	AUS	0.05451	0.091305	0.6682	A-M
186	Djibouti	DJI	0.144856	0.061307	4.6904	A-B
187	Kiribati	KIR	0.138971	0.062598	4.5826	A-B
188	Solomon Islands	SLB	0.130698	0.066662	4.6904	A-B
189	Bhutan	BTN	0	0	0	A-U
190	British Virgin Islands	VGB	0	0	0	A-U
191	Comoros	COM	0	0	0	A-U
192	East Timor	TLS	0	0	0	A-U
193	Gambia	GMB	0	0	0	A-U
194	Grenada	GRD	0	0	0	A-U
195	Guinea-Bissau	GNB	0	0	0	A-U
196	Lesotho	LSO	0	0	0	A-U
197	Montserrat	MSR	0	0	0	A-U
198	Nauru	NRU	0	0	0	A-U
199	Saint Helena	SHN	0	0	0	A-U
200	Saint Pierre and Miquelon	SPM	0	0	0	A-U
201	Turks and Caicos Islands	TCA	0	0	0	A-U

Gov Efficiency	Parallel Waste	GDP/cap ($)	Years	N
0.3137	0.4487	46501	1990-2025	36
0.2993	0.5507	38581	2000-2024	25
0.275	0.5871	44318	2000-2024	25
0.2747	0.6189		2000-2023	24
0.2702	0.5412	53755	2000-2024	25
0.2609	0.5664	40083	1990-2025	36
0.2603	0.5325	15035	2000-2024	25
0.2596	0.6975	4772	2000-2024	25
0.2544	0.5178	18709	2000-2024	25
0.2494	0.5767	5249	2000-2024	25
0.2408	0.5753	47943	1990-2025	36
0.2393	0.5902	25766	2000-2024	25
0.2384	0.6011	32774	1990-2025	23
0.2376	0.649	18999	2000-2024	25
0.2354	0.5795	32327	1990-2025	35
0.23	0.581	20610	1990-2025	32
0.2264	0.6326	15539	2000-2024	25
0.2257	0.649	41331	2000-2024	25
0.2211	0.5258	27464	1990-2025	33
0.2156	0.622	7220	2000-2024	25
0.2122	0.6269	19408	2000-2024	25
0.2093	0.621	17170	2000-2024	25
0.206	0.7193	8442	2000-2023	24
0.2053	0.6591	26694	1990-2025	31
0.2052	0.7107	3229	2000-2022	23
0.2023	0.662		2000-2023	24
0.2021	0.6076	7532	1990-2022	33
0.2013	0.697	24457	2000-2024	25
0.2012	0.6677	56432	2000-2024	25
0.1983	0.7144	8179	2000-2022	23
0.1983	0.5789	37601	1990-2025	36
0.1946	0.6845	1526	2000-2022	23
0.1932	0.5859	40217	1990-2025	36
0.1918	0.663	33935	1990-2025	36
0.1902	0.5985	41644	1990-2025	36
0.1887	0.5957	29628	1990-2025	35
0.188	0.7465	863	2000-2023	24
0.1868	0.7607		2000-2023	24
0.1839	0.7064	33544	1990-2025	24
0.1817	0.7465	3215	2000-2022	23
0.1814	0.7237	16966	2000-2024	25
0.181	0.7071	2872	2000-2023	24
0.18	0.6637	35129	2000-2024	25

0.1799	0.5781	736	2000-2022	23
0.1788	0.7029	10147	2000-2023	24
0.1788	0.7683	33580	2000-2024	25
0.1771	0.7652	3304	2000-2023	24
0.1756	0.7474	15843	2000-2023	24
0.175	0.7549	935	2000-2023	24
0.1742	0.7649	3898	2000-2023	24
0.1739	0.7659	12070	2000-2023	24
0.1739	0.7224	21191	2000-2023	24
0.1736	0.7265	12907	2000-2023	24
0.1728	0.6663	12347	2000-2024	25
0.1727	0.7687	1285	2000-2023	24
0.1699	0.7742	1621	2000-2023	24
0.1694	0.7012	17467	1990-2025	24
0.1681	0.7456	7377	2000-2023	24
0.1674	0.6793	21253	2000-2023	24
0.1668	0.7677	1286	2000-2023	24
0.1666	0.725	85777	1990-2025	36
0.1656	0.7151	53094	1990-2025	36
0.1649	0.7321		2000-2022	23
0.164	0.7606	47925	2000-2023	24
0.1639	0.7315	1381	2000-2023	24
0.1633	0.6629	15862	1990-2025	22
0.1632	0.6994	3646	2000-2022	23
0.1631	0.7666	7798	2000-2023	24
0.1631	0.746	7851	2000-2024	25
0.1631	0.648	46376	1990-2025	36
0.1627	0.7539		2000-2023	24
0.1625	0.7504		2000-2023	24
0.1625	0.759	2331	2000-2022	23
0.1623	0.7117	8068	2000-2023	24
0.1608	0.757	10545	2000-2024	25
0.1606	0.6893	9985	2000-2024	25
0.1602	0.7573	9651	2000-2024	25
0.1597	0.7657	9351	2000-2024	25
0.1593	0.7693	24960	2000-2023	24
0.1592	0.7443	1287	2000-2022	23
0.159	0.7128	12865	2000-2023	24
0.1584	0.7468	4802	2000-2023	24
0.1583	0.7388	7051	2000-2022	23
0.1582	0.7751	18523	2000-2024	25
0.158	0.7535		2000-2023	24
0.1577	0.7577		2000-2022	23
0.1571	0.7747	15662	2000-2024	25
0.1568	0.7749		2000-2022	23

0.1567	0.7849	947	2000-2023	24
0.1564	0.7753	1585	2000-2023	24
0.1561	0.7975		2000-2023	24
0.1554	0.7074	8184	2000-2024	25
0.1553	0.734	6297	2000-2024	25
0.155	0.7822		2000-2022	23
0.155	0.7612	27649	1990-2025	36
0.1549	0.703	11304	2000-2024	25
0.1544	0.7461	5340	2000-2024	25
0.1542	0.766	12893	2000-2024	25
0.154	0.7694	10929	2000-2023	24
0.1529	0.7802	4876	2000-2024	25
0.1528	0.7461		2000-2023	24
0.1512	0.757	5289	2000-2023	24
0.1504	0.7174	38474	2000-2024	25
0.1502	0.781	4309	2000-2023	24
0.1464	0.7177	16730	2000-2024	25
0.1458	0.7804	1538	2000-2022	23
0.1435	0.7772	11039	2000-2023	24
0.1434	0.7709	8820	2000-2022	23
0.1424	0.7679	2156	2000-2022	23
0.1421	0.762	4673	2000-2023	24
0.1411	0.7649	1268	2000-2022	23
0.1406	0.7535	49884	1990-2025	36
0.1406	0.7801	1533	2000-2023	24
0.1401	0.7665	29757	1990-2025	34
0.1399	0.7759	2040	2000-2023	24
0.1397	0.7471	32276	1990-2025	36
0.1392	0.7742	11706	2000-2023	24
0.1388	0.8052	44373	2000-2024	25
0.1385	0.7434	14044	2000-2024	25
0.1384	0.7288	15685	2000-2024	25
0.1378	0.7301	58099	1990-2025	36
0.1374	0.7678	2916	2000-2023	24
0.1374	0.7397	29968	1990-2025	32
0.1366	0.7337	26552	1990-2025	36
0.1363	0.7373	7169	2000-2023	24
0.1363	0.7474	8256	2000-2024	25
0.1357	0.7824	2975	2000-2023	24
0.1352	0.7704	4944	2000-2024	25
0.1343	0.7851	2098	2000-2023	24
0.1337	0.7882	4412	2000-2024	25
0.1333	0.8068		2000-2022	23
0.1322	0.7764	3830	2000-2023	24
0.1304	0.7223	8760	2000-2024	25

0.1298	0.7868	4304	2000-2023	24
0.1294	0.7898	3693	2000-2024	25
0.1291	0.7147	63623	2000-2022	23
0.1278	0.7969		2000-2023	24
0.1271	0.7951	3373	2000-2024	25
0.126	0.7534	36378	2000-2023	24
0.1235	0.8097	1644	2000-2023	24
0.1227	0.7885	20394	2000-2024	25
0.1213	0.7727	7637	2000-2023	24
0.1212	0.7623	37441	2000-2023	24
0.1175	0.7797	43761	1990-2025	36
0.1174	0.8038	13357	2000-2024	25
0.1172	0.7464	23862	2000-2023	24
0.1158	0.7851	24681	2000-2024	25
0.1155	0.7743	26580	1990-2025	32
0.1139	0.7536	27211	1990-2025	36
0.1127	0.7937		2000-2023	24
0.1111	0.7316	22684	2000-2024	25
0.1111	0.822	12517	2000-2023	24
0.1101	0.765	36114	1990-2025	36
0.1089	0.7841	41334	2000-2023	24
0.1009	0.776	670	2000-2023	24
0.0989	0.7842	24990	1990-2025	32
0.0961	0.7435		2000-2023	24
0.0888	0.8082	2359	2000-2023	24
0.0861	0.8257		2000-2022	23
0.0851	0.7947	28402	1990-2025	36
0.0849	0.8409	12532	2000-2024	25
0.0826	0.8232	9516	2000-2023	24
0.0813	0.8365	1545	2000-2023	24
0.0806	0.7657		2000-2023	24
0.0806	0.8756	14502	2000-2023	24
0.0781	0.8297	125238	2000-2024	25
0.0751	0.7648		2000-2022	23
0.071	0.8354	1307	2000-2022	23
0.0705	0.8302	4538	2000-2022	23
0.0698	0.8389		2000-2023	24
0.0676	0.8382		2000-2022	23
0.0674	0.6864		2000-2022	23
0.0652	0.687		2000-2023	24
0.0644	0.8514		2000-2023	24
0.0593	0.7143		2000-2023	24
0.0557	0.845		2000-2023	24
0.0542	0.836		2000-2023	24
0.0491	0.8555	770	2000-2023	24

0.0465	0.859		2000-2023	24
0.0454	0.8551		2000-2022	23
0.0397	0.8689	9783	2000-2022	23
0.0371	0.8595		2000-2022	23
0.0359	0.8624		2000-2023	24
0.0342	0.8706	13240	2000-2023	24
0.0327	0.9455	50322	2000-2024	25
0.0255	0.8551	2945	2000-2023	24
0.0249	0.861		2000-2022	23
0.023	0.8693		2000-2023	24
0	1		2000-2023	24
0	1		2000-2023	24
0	1	1978	2000-2023	24
0	1		2003-2023	21
0	1	1726	2000-2023	24
0	1		2000-2022	23
0	1	1548	2000-2022	23
0	1	2083	2000-2022	23
0	1		2000-2022	23
0	1		2000-2023	24
0	1		2000-2022	23
0	1		2000-2022	23
0	1		2000-2023	24

Rank	Country	K1 Geodesic Distance	K1 Entropy Gap	χ toward K1
1	Brazil	2.9029	0.3108	0.6454
2	Belarus	6.4394	0.9278	0.3343
3	Japan	5.1713	0.1156	0.5034
4	Finland	3.7773	0.2899	0.6447
5	Namibia	21.485	1.0129	0.2742
6	Argentina	4.5558	0.3465	0.4355
7	Democratic Republic of Cor	20.3762	1.3432	0.3159
8	Ukraine	5.0082	0.444	0.4307
9	Pakistan	5.1181	-0.1127	0.397
10	Zambia	21.269	1.3385	0.2214
11	Spain	3.3795	-0.0097	0.3009
12	Botswana	23.5566	1.57	0.3569
13	France	3.1137	0.6334	0.2791
14	Iran	9.2505	1.1833	0.3006
15	Bulgaria	3.4855	0.1263	0.2982
16	Uruguay	21.6969	0.4427	0.1868
17	Ethiopia	18.5748	1.6012	0.2879
18	Kazakhstan	21.0099	0.5835	0.2316
19	Senegal	16.3043	0.7475	0.1894
20	Seychelles	22.7648	1.2987	0.3398
21	Vanuatu	23.7374	1.023	0.1819
22	Mexico	5.01	0.4075	0.3373
23	Guatemala	20.2472	0.3641	0.2061
24	Iceland	17.7171	1.7528	0.273
25	Panama	16.521	0.3055	0.2015
26	China	4.2067	0.3661	0.6
27	Latvia	19.5106	0.4068	0.1726
28	Moldova	19.3156	1.0505	0.199
29	Indonesia	16.8468	0.6824	0.2323
30	Nepal	18.6332	1.6982	0.3614
31	Laos	21.3589	1.1988	0.2028
32	United Kingdom	4.6309	0.1015	0.2734
33	Israel	20.6068	0.9101	0.1395
34	Macao	25.3654	0.845	0.1828
35	Canada	4.224	0.3556	0.437
36	Costa Rica	20.7463	1.0829	0.2166
37	Honduras	22.0622	0.3931	0.1434
38	Antigua and Barbuda	21.9831	1.5441	0.1737
39	Hungary	4.7279	0.2953	0.2924
40	Taiwan	5.9617	0.4646	0.3209
41	Romania	3.9282	-0.0404	0.3148
42	Russia	8.2153	0.3889	0.5073
43	Eswatini	22.8979	0.7807	0.2245

44	Netherlands	6.9062	0.145	0.3172
45	Belize	23.5506	0.7285	0.1675
46	South Korea	6.2303	0.3	0.4787
47	Paraguay	19.0953	1.725	0.2846
48	Guyana	23.5022	1.4334	0.2315
49	United States	4.4681	0.1432	0.4119
50	Angola	22.5884	0.9372	0.2089
51	Ireland	15.9621	0.5677	0.2304
52	Azerbaijan	18.4899	1.3014	0.1851
53	Mozambique	21.6537	1.2243	0.147
54	Czechia	5.2189	0.3656	0.2819
55	Sao Tome and Principe	22.6343	1.4098	0.1963
56	Malawi	22.2418	1.4215	0.1576
57	Sweden	13.5276	0.3814	0.2027
58	Estonia	19.8328	0.3312	0.1209
59	Niger	25.7602	0.8999	0.2079
60	Georgia	22.0245	1.2236	0.1847
61	Fiji	23.4872	0.7743	0.1473
62	India	5.0513	0.7739	0.2675
63	Uganda	22.6965	1.2392	0.1584
64	Morocco	16.6347	0.566	0.1695
65	Colombia	21.6281	0.5203	0.1266
66	Chad	22.9289	1.5208	0.3358
67	Samoa	23.8961	0.6811	0.1632
68	Zimbabwe	22.2484	1.0359	0.1542
69	Jamaica	19.8623	0.6303	0.1312
70	Congo	22.272	1.0242	0.1435
71	Jordan	21.9939	0.9133	0.1411
72	Germany	16.4335	-0.0111	0.1842
73	Algeria	19.8041	1.6769	0.1777
74	Malta	23.3275	1.2173	0.1803
75	Norway	14.8895	1.369	0.1813
76	Mauritius	20.375	0.557	0.1104
77	Mauritania	23.1482	0.8821	0.1736
78	Slovenia	13.9177	0.2607	0.1279
79	Faroe Islands	24.0163	0.7589	0.1236
80	Guinea	22.9891	1.1527	0.2157
81	Saint Kitts and Nevis	21.8994	1.5738	0.1119
82	Togo	24.4849	0.8581	0.1317
83	Rwanda	24.4544	0.5194	0.1094
84	Suriname	22.1976	0.9762	0.1611
85	Lebanon	21.2673	0.7222	0.091
86	Serbia	16.2651	0.7607	0.1219
87	Gibraltar	18.4957	1.7587	0.0743
88	Benin	24.5843	1.0812	0.1588

89	Bolivia	19.4466	0.6964	0.0918
90	Ghana	22.0199	0.9743	0.168
91	Burkina Faso	23.3945	1.1086	0.1719
92	Eritrea	22.2739	1.4046	0.1794
93	Burundi	23.9249	1.0318	0.0954
94	South Africa	14.2403	1.082	0.0673
95	Cote d'Ivoire	21.5391	1.0779	0.1428
96	Sierra Leone	23.0133	1.2576	0.1477
97	Vietnam	16.1346	0.4894	0.1189
98	Uzbekistan	23.6639	1.0772	0.1672
99	Tajikistan	23.4499	1.4259	0.1591
100	Bahrain	19.7556	1.7413	0.181
101	Maldives	22.0823	1.5035	0.151
102	American Samoa	19.5593	1.7587	0.0683
103	Trinidad and Tobago	21.9174	1.7232	0.1877
104	United States Virgin Islands	21.7236	1.6244	0.1506
105	Austria	19.7633	0.3397	0.1102
106	Croatia	19.3708	0.1991	0.0843
107	Aruba	22.9853	1.2236	0.1455
108	Belgium	14.2078	0.1084	0.0721
109	Portugal	19.7961	0.1643	0.0857
110	El Salvador	21.7365	0.4721	0.1048
111	Brunei	25.52	1.1874	0.1252
112	Puerto Rico	17.2312	0.5586	0.0954
113	Syria	22.5822	0.913	0.1107
114	Bangladesh	24.2212	0.749	0.0855
115	Papua New Guinea	22.7524	0.7669	0.1296
116	Venezuela	21.5492	1.0972	0.1391
117	Slovakia	16.2095	0.6111	0.0872
118	Guam	22.1221	1.4854	0.1083
119	Nicaragua	21.9424	0.4193	0.1004
120	Cameroon	22.615	0.8267	0.1109
121	Philippines	16.4138	0.7062	0.1537
122	Saint Lucia	21.6499	1.6418	0.1077
123	New Caledonia	23.2231	0.4684	0.0899
124	Oman	22.7987	1.4497	0.0568
125	Central African Republic	16.8166	1.7587	0.0668
126	Cambodia	22.9426	0.7633	0.1226
127	North Korea	23.9564	0.986	0.1105
128	Somalia	22.9649	1.2	0.0458
129	Madagascar	23.5473	0.5627	0.0771
130	Liberia	22.9995	1.1222	0.0778
131	Kuwait	23.5849	0.9696	0.0803
132	Denmark	16.2082	0.5383	0.1267
133	Myanmar	21.0871	0.7196	0.0907

134	Barbados	24.1298	1.3527	0.0802
135	Tanzania	21.7801	1.1137	0.1433
136	Albania	19.3472	1.6126	0.0685
137	Ecuador	19.296	0.9648	0.1919
138	Hong Kong	23.3906	1.0213	0.0749
139	Falkland Islands	19.5593	1.7587	0.054
140	Turkmenistan	19.324	1.756	0.0848
141	Thailand	16.1381	0.7234	0.0822
142	Italy	16.135	0.2784	0.1327
143	Tonga	22.3611	1.3486	0.0592
144	Cayman Islands	21.7054	1.6289	0.0635
145	Yemen	25.0118	0.8707	0.0585
146	Tunisia	20.8516	1.5061	0.0765
147	Turkey	16.2611	0.1566	0.073
148	Chile	16.3168	0.0041	0.1075
149	Iraq	23.7278	1.0194	0.1324
150	Bahamas	21.2676	1.7038	0.0593
151	Haiti	22.8363	1.2752	0.29
152	Afghanistan	24.328	0.9974	0.2032
153	Kyrgyzstan	25.0368	1.1887	0.1154
154	Luxembourg	19.6304	0.3	0.1082
155	Poland	16.5189	0.2974	0.0771
156	New Zealand	16.0568	0.6267	0.0892
157	Cook Islands	22.6281	1.0655	0.039
158	Egypt	22.2652	1.0378	0.0764
159	Peru	15.7724	0.6939	0.0625
160	North Macedonia	16.449	0.259	0.0493
161	Dominican Republic	16.8634	0.1773	0.0744
162	Cyprus	23.1433	1.0707	0.0435
163	Lithuania	19.8476	0.2566	0.0717
164	Greece	16.3908	0.1798	0.088
165	Greenland	22.4679	1.373	0.4099
166	Armenia	22.9229	0.4812	0.0358
167	Mali	23.3447	0.8897	0.1818
168	Switzerland	13.6246	0.6761	0.0754
169	Kenya	21.7979	0.4668	0.05
170	Mongolia	22.1873	1.1956	0.104
171	Dominica	22.7074	1.366	0.2126
172	Cape Verde	23.4447	0.9716	0.1446
173	Malaysia	21.3606	0.5805	0.0658
174	Djibouti	22.6193	1.1112	0.0215
175	Sudan	22.7179	1.0666	0.0909
176	Kiribati	22.5458	1.1963	0.0207
177	Palestine	22.6115	1.1222	0.0192
178	Solomon Islands	22.1854	1.454	0.0197

179	Equatorial Guinea	24.104	1.1017	0.2946
180	Cuba	19.4293	1.1277	0.0406
181	Gabon	24.7247	0.7192	0.128
182	Qatar	19.8451	1.7413	0.2342
183	Saint Vincent and the Grena	22.7074	1.366	0.1258
184	Libya	23.5538	1.2059	0.1038
185	Nigeria	21.3218	1.1963	0.045
186	Australia	16.4198	0.2623	0.0303
187	French Polynesia	23.77	0.946	0.055
188	Bermuda	21.9955	1.6782	0.1853
189	Bhutan	16.8166	1.7587	0
190	British Virgin Islands	19.5593	1.7587	0
191	Comoros	19.5593	1.7587	0
192	East Timor	19.5593	1.7587	0
193	Gambia	19.5593	1.7587	0
194	Grenada	19.5593	1.7587	0
195	Guinea-Bissau	19.5593	1.7587	0
196	Lesotho	16.8166	1.7587	0
197	Montserrat	19.5593	1.7587	0
198	Nauru	19.5593	1.7587	0
199	Saint Helena	19.5593	1.7587	0
200	Saint Pierre and Miquelon	19.5593	1.7587	0
201	Turks and Caicos Islands	19.5593	1.7587	0

Avg Speed	Effective Speed	ETA K1 (years)	Fossil Share %	Low Carbon %
0.9315	0.6013	4.8	10.55087757	89.4491272
3.3149	1.108	5.8	62.50820923	37.49179077
1.738	0.875	5.9	68.71083069	31.28916931
0.9149	0.5898	6.4	3.664084435	96.33591461
9.2579	2.5383	8.5	2.116402149	97.88359833
0.9645	0.4201	10.8	58.76336288	41.23664093
5.8754	1.856	11	0	100
0.9838	0.4237	11.8	27.82710075	72.17289734
0.9139	0.3628	14.1	53.64085388	46.3591423
6.5142	1.442	14.7	10.99691772	89.00308228
0.7538	0.2268	14.9	25.2979908	74.70200348
4.1738	1.4896	15.8	99.61389923	0.386100382
0.6269	0.1749	17.8	5.16443491	94.83557129
1.7118	0.5146	18	92.82447815	7.175522327
0.643	0.1917	18.2	28.06001472	71.93997955
6.348	1.1858	18.3	5.385060787	94.61493683
3.4641	0.9972	18.6	0	100
4.8629	1.1264	18.7	84.78609467	15.21390629
4.6083	0.8729	18.7	78.26087189	21.73913002
3.5189	1.1958	19	85.7142868	14.2857151
6.6254	1.2051	19.7	75	25
0.7466	0.2518	19.9	75.41809845	24.58190155
4.8735	1.0043	20.2	25.48291206	74.51708221
3.1786	0.8677	20.4	0	100
3.9531	0.7965	20.7	38.22843933	61.77156067
0.3316	0.199	21.1	61.83478165	38.16522217
5.1774	0.8934	21.8	26.99680519	73.00319672
4.4463	0.8849	21.8	88.29174042	11.70825386
3.0033	0.6976	24.2	81.91042328	18.08957291
2.1209	0.7665	24.3	0	100
4.2597	0.8639	24.7	23.28871918	76.71128082
0.6706	0.1834	25.3	35.52496719	64.47503662
5.7235	0.7987	25.8	89.49065399	10.50934029
5.3345	0.975	26	59.18367004	40.81632614
0.3596	0.1572	26.9	22.30562592	77.69437408
3.482	0.7541	27.5	5.97738266	94.02262115
5.3686	0.7698	28.7	38.42869186	61.57130814
4.246	0.7376	29.8	94.44444275	5.555555344
0.54	0.1579	29.9	24.53669357	75.46330261
0.6208	0.1992	29.9	84.85929871	15.14069843
0.4035	0.127	30.9	32.48343658	67.51656342
0.5144	0.261	31.5	64.0646286	35.93536758
3.215	0.7216	31.7	3.571428537	96.42857361

0.6865	0.2178	31.7	45.78215027	54.21784592
4.3097	0.7219	32.6	11.11111164	88.88889313
0.3981	0.1905	32.7	60.10905457	39.89094543
1.9732	0.5616	34	0	100
2.965	0.6865	34.2	93.28357697	6.716417789
0.3102	0.1278	35	58.1306839	41.8693161
3.0704	0.6415	35.2	23.57859421	76.42140198
1.9601	0.4517	35.3	51.62955856	48.37044144
2.8117	0.5206	35.5	88.04425049	11.9557457
4.0973	0.6023	36	16.30879402	83.69121552
0.5024	0.1416	36.8	40.8422699	59.15772247
3.0757	0.6036	37.5	88.8888855	11.11111069
3.6854	0.5809	38.3	4.371584415	95.62841034
1.6778	0.3401	39.8	1.218654752	98.78135681
4.0576	0.4906	40.4	40.49586868	59.5041275
3.0587	0.6359	40.5	97.49999237	2.5
2.8763	0.5313	41.5	19.8457222	80.15428162
3.8074	0.5608	41.9	36.52173996	63.47826385
0.4478	0.1198	42.2	77.74752045	22.25248718
3.323	0.5265	43.1	2.608695745	97.39130402
2.2426	0.3802	43.8	73.49589539	26.5041008
3.8599	0.4886	44.3	35.64163971	64.35836029
1.4763	0.4958	46.2	94.87180328	5.128205299
3.1215	0.5095	46.9	60	39.99999619
3.0512	0.4704	47.3	32.49097443	67.50902557
3.1406	0.4119	48.2	87.08241272	12.91759491
3.1568	0.4531	49.2	79.30367279	20.6963253
3.1456	0.4439	49.5	76.87195587	23.12804604
1.757	0.3237	50.8	41.15672302	58.84327698
2.17	0.3856	51.4	99.05503082	0.944963634
2.4642	0.4443	52.5	82.11008453	17.88990784
1.5309	0.2776	53.6	1.207608461	98.79238892
3.0804	0.3402	59.9	82.56880951	17.4311924
2.2038	0.3827	60.5	72.56098175	27.43902397
1.7464	0.2233	62.3	21.03494644	78.96504974
3.0274	0.3742	64.2	54.16666412	45.83333588
1.6035	0.3458	66.5	25.18518448	74.81481171
2.8692	0.321	68.2	95.45454407	4.545454502
2.569	0.3384	72.3	79.3478241	20.652174
3.0706	0.336	72.8	43.39622879	56.60377884
1.8873	0.304	73	57.00934601	42.99065399
3.1929	0.2907	73.2	52.65486908	47.34513474
1.8231	0.2222	73.2	72.97953033	27.02047348
3.3274	0.2473	74.8	100	0
2.0671	0.3282	74.9	97	3

2.8129	0.2583	75.3	62.02428818	37.97570801
1.6871	0.2835	77.7	61.46562195	38.53437424
1.7342	0.2982	78.5	82.65895844	17.34104156
1.5447	0.2771	80.4	88.63636017	11.36363697
3.0713	0.2929	81.7	30.7692318	69.23077393
2.554	0.1719	82.8	83.81177521	16.18821907
1.791	0.2557	84.2	68.91284943	31.08715248
1.8412	0.272	84.6	4.761904716	95.23809814
1.5585	0.1853	87.1	57.71790314	42.28210068
1.5565	0.2603	90.9	90.75409698	9.245902061
1.6001	0.2546	92.1	8.579336166	91.42066193
1.1789	0.2133	92.6	99.75118256	0.248825014
1.5214	0.2298	96.1	92.94117737	7.058823109
2.907	0.1984	98.6	100	0
1.155	0.2168	101.1	99.89473724	0.105263159
1.4185	0.2137	101.7	97.0149231	2.98507452
1.6685	0.1839	107.4	15.87471485	84.12528229
2.1039	0.1774	109.2	24.01927185	75.98072815
1.4428	0.2099	109.5	83	17
1.7932	0.1293	109.9	27.97969818	72.02029419
2.0836	0.1786	110.9	19.038311	80.96168518
1.848	0.1937	112.2	6.808510303	93.19149017
1.6543	0.2071	123.2	100	0
1.4436	0.1378	125.1	94.17735291	5.822649956
1.5986	0.1769	127.7	95.61753082	4.382470131
2.2108	0.1891	128.1	97.97119141	2.028807163
1.3684	0.1773	128.3	76.3213501	23.67864609
1.1891	0.1654	130.3	21.61803627	78.38196564
1.3956	0.1217	133.2	14.84536076	85.1546402
1.5209	0.1647	134.3	92.22222137	7.777778149
1.5917	0.1599	137.3	35.46910858	64.53089142
1.4238	0.1579	143.2	36.13445282	63.86554718
0.7422	0.1141	143.9	78.26950836	21.73049355
1.3925	0.15	144.3	97.49999237	2.5
1.6949	0.1523	152.5	73.78640747	26.21359253
2.4873	0.1414	161.3	95.80161285	4.198385715
1.558	0.1041	161.5	0	100
1.1526	0.1413	162.4	59.28752899	40.71246719
1.3306	0.1471	162.9	36.91149139	63.08851242
2.9494	0.1352	169.9	80.95238495	19.04761887
1.7734	0.1368	172.1	64.77272797	35.22727203
1.6523	0.1285	179	66.66666412	33.33333206
1.5961	0.1282	183.9	97.82112122	2.178872824
0.6718	0.0852	190.3	8.797127724	91.20287323
1.2072	0.1095	192.6	59.76499176	40.23500824

1.5602	0.1251	193	91.81817627	8.181818008
0.7805	0.1119	194.7	74.4990921	25.50091171
1.449	0.0992	194.9	0	100
0.4734	0.0908	212.4	28.33333397	71.66666412
1.4621	0.1095	213.7	99.04357147	0.956429362
1.6688	0.0901	217	100	0
0.9956	0.0844	229	99.97039795	0.029594554
0.8517	0.07	230.5	85.15938568	14.84061718
0.5089	0.0675	238.9	51.31708145	48.68291855
1.569	0.0929	240.7	85.7142868	14.28571415
1.3936	0.0886	245.1	97.14286041	2.857142925
1.7103	0.1	250.1	83.06188965	16.93811035
1.0516	0.0804	259.3	96.0153656	3.984637499
0.8149	0.0594	273.5	56.74277878	43.25722504
0.554	0.0596	274	30.10306931	69.89692688
0.6535	0.0865	274.3	98.82893372	1.171069264
1.2911	0.0765	277.9	99.02439117	0.975609779
0.2831	0.0821	278.1	81.18811798	18.81188202
0.4254	0.0864	281.5	13.40206051	86.59793091
0.7551	0.0871	287.4	17.43684578	82.56315613
0.6062	0.0656	299.4	8.496731758	91.50326538
0.663	0.0511	323	68.91805267	31.08194351
0.5546	0.0495	324.5	14.93477249	85.06523132
1.7797	0.0695	325.8	50	50
0.8687	0.0664	335.4	88.80323792	11.19676495
0.736	0.046	343.1	40.72043991	59.27956009
0.9616	0.0474	346.7	63.4974556	36.50254822
0.6472	0.0482	350.1	81.45720673	18.54279137
1.5105	0.0657	352.2	72.29965973	27.70034981
0.7729	0.0554	358.1	22.61363602	77.38636017
0.5046	0.0444	369.1	50.38679886	49.61320496
0.1468	0.0602	373.3	12.96296215	87.03703308
1.6036	0.0575	399	39.595047	60.40494537
0.2892	0.0526	443.8	57.33944702	42.66054916
0.4035	0.0304	447.9	2.575733185	97.424263
0.9297	0.0465	469.2	8.093384743	91.90661621
0.4044	0.042	527.8	90.33412933	9.665870667
0.1933	0.0411	552.5	86.66665649	13.33333206
0.2706	0.0391	599.2	72	28
0.5259	0.0346	617	80.32489014	19.67510796
1.6311	0.035	645.9	65	35
0.3858	0.0351	648	29.85074615	70.14925385
1.5975	0.0331	681.3	75	25
1.6297	0.0313	721.8	66.66667175	33.33333588
1.5001	0.0295	752	90.90909576	9.090909004

0.0931	0.0274	878.6	68.78981018	31.21018982
0.5335	0.0217	896.1	95.29103851	4.708960533
0.1877	0.024	1028.9	52.03761673	47.96237946
0.081	0.019	1046.6	99.76125336	0.238744885
0.1518	0.0191	1189.4	86.66665649	13.33333206
0.1574	0.0163	1442.3	99.97151184	0.028481914
0.2823	0.0127	1679.4	76.97483063	23.02516747
0.3092	0.0094	1750.3	64.9233017	35.07669067
0.2174	0.012	1986.2	66.1971817	33.8028183
0.0266	0.0049	4464.7	98.4375	1.5625
0	0		0	100
0	0		100	0
0	0		100	0
0	0		100	0
0	0		100	0
0	0		100	0
0	0		100	0
0	0		0	100
0	0		100	0
0	0		100	0
0	0		100	0
0	0		100	0
0	0		100	0

Development Stage	Degradation Flag
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Post-fossil (K1 approaching)	
Mid-transition	
Post-fossil (K1 approaching)	
Mid-transition	
Mid-transition	
Post-fossil (K1 approaching)	
Mid-transition	
Early transition	
Post-fossil (K1 approaching)	
Degrading (moving away fro	YES
Active transition	
Post-fossil (K1 approaching)	YES
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Early transition	
Unstable (burst-driven, no clear path)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Mid-transition	YES
Early transition	
Early transition	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Active transition	
Early transition	
Early transition	
Active transition	YES
Post-fossil (K1 approaching)	YES
Early transition	
Unstable (burst-driven, no clear path)	
Active transition	
Early transition	
Mid-transition	
Early transition	
Post-fossil (K1 approaching)	

Mid-transition	
Post-fossil (K1 approaching)	YES
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Mid-transition	
Early transition	YES
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Mid-transition	
Early transition	
Post-fossil (K1 approaching)	YES
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Early transition	YES
Pre-transition (fossil locked)	
Early transition	
Early transition	
Early transition	
Degrading (moving away fro	YES
Early transition	
Mid-transition	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Degrading (moving away fro	YES
Early transition	
Mid-transition	
Early transition	YES
Early transition	
Unstable (burst-driven, no clear path)	
Degrading (moving away fro	YES
Early transition	
Early transition	YES
Early transition	
Early transition	
Unstable (burst-driven, no clear path)	
Pre-transition (fossil locked)	

Early transition	
Early transition	
Early transition	
Pre-transition (fossil locked)	
Early transition	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Pre-transition (fossil locked)	
Unstable (burst-driven, no clear path)	
Early transition	
Pre-transition (fossil locked)	
Post-fossil (K1 approaching)	
Early transition	
Pre-transition (fossil locked)	
Active transition	
Post-fossil (K1 approaching)	
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Early transition	
Early transition	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Pre-transition (fossil locked)	
Early transition	
Early transition	
Early transition	
Pre-transition (fossil locked)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	YES
Early transition	
Pre-transition (fossil locked)	
Degrading (moving away fr	YES
Unstable (burst-driven, no clear path)	
Pre-transition (fossil locked)	
Post-fossil (K1 approaching)	
Early transition	YES

Early transition	
Degrading (moving away fr	YES
Post-fossil (K1 approaching)	
Early transition	YES
Early transition	
Unstable (burst-driven, no clear path)	
Pre-transition (fossil locked)	
Early transition	
Early transition	
Pre-transition (fossil locked)	
Pre-transition (fossil locked)	
Pre-transition (fossil locked)	
Early transition	
Early transition	
Early transition	
Early transition	
Pre-transition (fossil locked)	
Early transition	
Post-fossil (K1 approaching)	
Post-fossil (K1 approaching	YES
Post-fossil (K1 approaching)	
Early transition	
Post-fossil (K1 approaching)	
Unstable (burst-driven, no clear path)	
Pre-transition (fossil locked)	
Mid-transition	
Early transition	
Early transition	
Early transition	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Early transition	
Early transition	
Post-fossil (K1 approaching)	
Post-fossil (K1 approaching)	
Early transition	
Degrading (moving away fr	YES
Early transition	
Early transition	
Unstable (burst-driven, no clear path)	
Early transition	
Unstable (burst-driven, no clear path)	
Early transition	
Pre-transition (fossil locked)	

Early transition
Early transition
Early transition
Pre-transition (fossil locked)
Early transition
Pre-transition (fossil locked)
Early transition
Early transition
Early transition
Early transition
Post-fossil (K1 approaching)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Post-fossil (K1 approaching)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)
Pre-transition (fossil locked)

Rank	Country	Stagnation Cost ($M/yr)	Dollar Waste (% GDP)
1	China	280112.7	107.1444
2	United States	84259.3	90.8164
3	India	71813.2	59.7209
4	Russia	26966.4	136.458
5	Japan	24567.7	57.631
6	South Korea	12992.6	109.9017
7	Iran	12837.1	137.8645
8	Indonesia	12637	52.3363
9	Taiwan	9164.5	57.1133
10	South Africa	8881.5	130.7872
11	Mexico	8580.1	71.4517
12	Turkey	8399.6	62.1941
13	Germany	8247.6	39.6585
14	Australia	7799	116.813
15	Vietnam	7351.5	115.9466
16	Egypt	6819	63.9228
17	Malaysia	5990.5	121.2625
18	Canada	5901.5	132.156
19	Thailand	5534	89.9769
20	Iraq	5180.5	99.7798
21	Poland	5098.5	72.0262
22	Kazakhstan	4752.5	113.3839
23	Uzbekistan	4274.5	117.0103
24	Brazil	3954.6	62.3449
25	Bangladesh	3950	45.9806
26	Philippines	3853.5	40.6406
27	Italy	3768.6	62.8727
28	Pakistan	3626	42.8126
29	United Kingdom	3178	44.4927
30	Algeria	3051	89.5284
31	Kuwait	2793	172.3399
32	Argentina	2629.5	60.6369
33	Spain	2211.5	64.9531
34	Turkmenistan	2207	421.2258
35	Israel	2110.5	68.8807
36	Qatar	1767.5	122.7965
37	Netherlands	1711	62.3991
38	Bahrain	1632	321.8011
39	Czechia	1515.5	75.8232
40	Libya	1458	197.1385
41	Ukraine	1396.5	121.6031
42	Serbia	1292	107.0225
43	Hong Kong	1283.5	46.3626

44	Morocco	1265	60.5505
45	Colombia	1250.5	62.7799
46	France	1187.5	48.8948
47	Chile	1183.5	82.1627
48	Oman	1182	183.9088
49	Nigeria	1019	35.601
50	Greece	918	87.3884
51	Azerbaijan	886.5	86.7837
52	Peru	840.5	68.2033
53	Venezuela	747.5	326.7517
54	Belarus	738.5	108.2424
55	Dominican Republic	734	49.7492
56	Myanmar	703	34.3028
57	Syria	685	133.9063
58	Romania	624.5	47.0275
59	Puerto Rico	622	42.5939
60	Jordan	608.5	86.3837
61	Laos	607	86.7334
62	Tunisia	583.5	71.4882
63	Ghana	550	55.7844
64	Belgium	546	84.8731
65	Bulgaria	523	96.3063
66	Cuba	488.5	78.8572
67	North Korea	457	350.6375
68	Austria	425.5	74.6482
69	Ireland	396.5	44.1397
70	Cambodia	391.5	79.8119
71	Ecuador	346	96.3689
72	Mongolia	338.5	89.554
73	Hungary	328.5	55.7444
74	Portugal	327	70.562
75	Trinidad and Tobago	324	354.8999
76	Sweden	301.5	81.5139
77	Bolivia	289	97.0144
78	New Zealand	266	83.2194
79	Brunei	249.5	
80	Norway	232.5	80.2015
81	Finland	231	82.9557
82	Cote d'Ivoire	219	49.2916
83	Senegal	215.5	61.4977
84	Tanzania	204	30.325
85	Denmark	191	48.8796
86	Congo	184.5	42.7957
87	Guatemala	183.5	46.5058
88	Sudan	179.5	

89	Honduras	169.5	81.2222
90	Panama	166.5	73.4516
91	Moldova	164	116.6969
92	North Macedonia	156.5	70.8141
93	Angola	150	47.6966
94	Cyprus	140	72.5159
95	Kyrgyzstan	140	130.9061
96	Slovakia	138	62.9075
97	Slovenia	136.5	61.1625
98	Jamaica	126	123.3565
99	Mozambique	125	134.9248
100	Zimbabwe	124	174.7456
101	Papua New Guinea	121.5	
102	Cameroon	119	33.8546
103	Croatia	116.5	70.5297
104	Switzerland	115	38.0059
105	Botswana	110	55.7011
106	Armenia	108	103.3134
107	Zambia	108	44.7453
108	Tajikistan	106	67.7969
109	Mauritius	103.5	57.5547
110	Georgia	102	74.4225
111	Estonia	96.5	121.3055
112	Yemen	90	46.9696
113	New Caledonia	87.5	
114	Mali	86	73.6596
115	Lebanon	83.5	105.7334
116	Uruguay	83.5	54.1444
117	Gabon	68.5	38.4148
118	Bahamas	67	
119	Madagascar	63	32.8319
120	Nicaragua	63	48.9458
121	Lithuania	61.5	56.7467
122	Guam	55	
123	Paraguay	55	66.0757
124	Kenya	54.5	36.4796
125	Malta	51.5	146.4937
126	Burkina Faso	48	47.2444
127	Equatorial Guinea	47.5	108.0258
128	Latvia	43.5	57.2064
129	Guyana	42.5	
130	Suriname	41	
131	Mauritania	39.5	96.586
132	Costa Rica	39	44.7587
133	Guinea	37	73.665

134	El Salvador	35	55.7172
135	Barbados	33	126.0023
136	Benin	29.5	61.509
137	Iceland	28.5	311.885
138	Aruba	27.5	
139	Haiti	27	47.0669
140	Niger	27	42.7361
141	Maldives	26	
142	Palestine	23.5	80.1133
143	Cayman Islands	22.5	
144	Togo	22	64.3931
145	Democratic Republic of Con	21.5	30.4177
146	Ethiopia	21.5	23.4239
147	United States Virgin Islands	21.5	
148	Bermuda	20.5	
149	Seychelles	18	88.1575
150	East Timor	17	
151	Gambia	17	45.0854
152	Uganda	16.5	23.6235
153	Fiji	16	
154	Rwanda	16	18.5257
155	French Polynesia	15.5	
156	Bhutan	13.5	
157	Eritrea	13	
158	Saint Lucia	13	125.873
159	Nepal	12.5	41.23
160	Cape Verde	12	50.1406
161	Chad	12	28.6424
162	Albania	11	50.9667
163	Antigua and Barbuda	11	
164	Macao	11	
165	Somalia	11	
166	Luxembourg	9.5	76.8454
167	Faroe Islands	8.5	
168	Liberia	8.5	57.6984
169	Turks and Caicos Islands	8.5	
170	Grenada	8	
171	Saint Kitts and Nevis	7	
172	Gibraltar	6.5	
173	Afghanistan	6	64.963
174	American Samoa	5.5	
175	British Virgin Islands	5.5	
176	Malawi	5	30.9958
177	Burundi	4.5	30.7918
178	Comoros	4.5	102.567

179	Djibouti	4.5	83.7811
180	Dominica	4.5	74.9618
181	Namibia	4.5	61.325
182	Saint Vincent and the Grena	4.5	
183	Eswatini	4	50.008
184	Belize	3.5	
185	Solomon Islands	3.5	
186	Greenland	3	
187	Samoa	3	
188	Guinea-Bissau	2.5	44.2665
189	Sao Tome and Principe	2.5	49.3171
190	Tonga	2	
191	Vanuatu	2	
192	Nauru	1.5	
193	Saint Pierre and Miquelon	1.5	
194	Kiribati	1	
195	Cook Islands	0.5	
196	Falkland Islands	0.5	
197	Lesotho	0.5	101.2285
198	Montserrat	0.5	
199	Saint Helena	0.5	
200	Sierra Leone	0.5	34.6974
201	Central African Republic		21.6368

Parallel Waste	Energy/GDP	Carbon Intensity (gCO2/kWh)	Elec Gen (TWh)
0.649	1.650836706	555.4000244	10086.87988
0.6677	1.360092282	383.7799988	4391.02002
0.622	0.960217535	707.4500122	2030.199951
0.5902	2.311911106	445.980011	1209.310059
0.5507	1.046555042	483.4299927	1016.390015
0.649	1.69331944	415.5100098	625.3800049
0.621	2.219882727	648.6799927	395.7900085
0.6663	0.785476804	680.25	371.5400085
0.5412	1.055248499	635.1900024	288.5599976
0.703	1.860482097	713.1699829	249.0700073
0.6326	1.129546762	483.1400146	355.1799927
0.7536	0.825252652	474.7399902	353.8599854
0.4487	0.883834302	331.6000061	497.4400024
0.9455	1.23547554	553.8300171	281.6400146
0.7474	1.551317692	484.3200073	303.5799866
0.8409	0.760149658	574.5	237.3899994
0.7851	1.544591188	604.4299927	198.2200012
0.5871	2.250911713	185.3500061	636.789978
0.7288	1.234611154	554.7299805	199.5200043
0.7742	1.288767338	689.4000244	150.2899933
0.7471	0.964132905	592.2000122	172.1900024
0.697	1.626742244	802.039978	118.5100021
0.7772	1.505496383	1121.180054	76.25
0.5325	1.17076695	106.0599976	745.7199707
0.7704	0.596839249	693.8400269	113.8600006
0.7223	0.562645912	612.3499756	125.8600006
0.765	0.821843266	285.2600098	264.2200012
0.5767	0.74239862	398.2399902	182.1000061
0.5789	0.768553197	217.0899963	292.7799988
0.7265	1.232331395	633.6500244	96.30000305
0.8052	2.140293837	637.2399902	87.66000366
0.5178	1.171159983	344.8299866	152.5099945
0.663	0.979709804	153.2599945	288.6000061
0.8706	4.838528156	1306.300049	33.79000092
0.7534	0.914218485	567.2600098	74.41000366
0.8297	1.479926944	602.8300171	58.63999939
0.5753	1.084683776	253.2200012	135.1399994
0.7623	4.221349239	902.4099731	36.16999817
0.5795	1.308400273	401.3999939	75.51000214
0.822	2.398316145	830.5300293	35.11000061
0.6076	2.001447201	250.4700012	111.5100021
0.7012	1.526239872	696.1199951	37.11999893
0.7606	0.609517992	681.9899902	37.63999939

0.7074	0.855958998	576.5700073	43.88000107
0.7434	0.844503224	285.7999878	87.51000214
0.5664	0.863235056	41.79000092	568.3099976
0.7316	1.123048306	265.1799927	89.26000214
0.7683	2.393847942	545.3300171	43.34999847
0.7802	0.456333548	507.8500061	40.13000107
0.7337	1.191144228	315.6300049	58.16999817
0.7237	1.199096918	632.7600098	28.02000046
0.6893	0.989439309	263.269989	63.84999847
0.757	4.316129208	180.25	82.94000244
0.6269	1.726731539	323.269989	45.68999863
0.7177	0.693144023	565.9199829	25.94000053
0.7461	0.459732592	569.6900024	24.68000031
0.759	1.764137506	682.2700195	20.07999992
0.6591	0.713564515	250.75	49.81000137
0.6637	0.641746223	664.5300293	18.71999931
0.7709	1.120602846	539.210022	22.56999969
0.7373	1.176399589	232.1199951	52.29999924
0.757	0.944322646	560.25	20.82999992
0.7868	0.709015548	452.8599854	24.29000092
0.5985	1.418181539	149.7700043	72.91000366
0.581	1.65756166	275.3399963	37.99000168
0.7727	1.020480037	638.9799805	15.28999996
0.7804	4.492852211	344.2600098	26.54999924
0.7797	0.957393587	113.9100037	74.70999908
0.7301	0.604594886	255.8899994	30.98999977
0.7898	1.010572314	498.0899963	15.72000027
0.7573	1.272466063	209.6999969	33
0.766	1.169084907	807.8800049	8.380000114
0.5957	0.935742199	162.3399963	40.47000122
0.7947	0.887954772	127.8300018	51.15999985
0.7693	4.613193035	682.1099854	9.5
0.648	1.257917166	35.33000183	170.6799927
0.734	1.321749449	468.019989	12.35000038
0.7174	1.160052061	119.6600037	44.45999908
0.7975		892.6699829	5.590000153
0.725	1.106232643	29.10000038	159.8200073
0.5859	1.415983558	56.81000137	81.33000183
0.7649	0.644415557	393.5299988	11.13000011
0.7071	0.86975342	535.4000244	8.050000191
0.7824	0.387568742	371.5899963	10.97999954
0.7535	0.648663342	114.3000031	33.41999817
0.7764	0.55122906	713.7299805	5.170000076
0.7117	0.653466225	272.6600037	13.46000004
0.7969		214.3300018	16.75

0.762	1.065840721	289.5	11.71000004
0.6793	1.081363916	258.7399902	12.86999989
0.746	1.564399719	629.5599976	5.210000038
0.6629	1.0682621	531.4099731	5.889999866
0.781	0.610732615	167.2200012	17.94000053
0.7842	0.924665511	487.8099976	5.739999771
0.7882	1.66079998	172.5200043	16.22999954
0.5258	1.196411014	94.84999847	29.10000038
0.6011	1.017531157	183.4700012	14.88000011
0.7388	1.669795275	561.25	4.489999771
0.7549	1.787427306	127.8099976	19.55999947
0.7753	2.25391531	298.4400024	8.31000042
0.7539		513.7399902	4.730000019
0.7678	0.440944463	285.7099915	8.329999924
0.7743	0.910863221	160.3600006	14.52999973
0.7147	0.531807721	37.25999832	61.72999954
0.7474	0.745248735	849.4199829	2.589999914
0.8038	1.285336494	242.9700012	8.890000343
0.7652	0.584783375	111	19.45999908
0.6975	0.971951365	97.79000092	21.68000031
0.7224	0.796763361	633.0300293	3.269999981
0.7751	0.960160077	143.0599976	14.26000023
0.7665	1.582663536	319.0100098	6.050000191
0.8365	0.561509311	586.3200073	3.069999933
0.7749		566.3400269	3.089999914
0.7687	0.958179832	394.5	4.360000134
0.7193	1.470034719	369.4700012	4.519999981
0.7885	0.686672926	96.69999695	17.27000046
0.8756	0.438732445	429.4700012	3.190000057
0.8595		653.6599731	2.049999952
0.7649	0.429206699	477.269989	2.640000105
0.7468	0.655365586	288.3299866	4.369999886
0.7397	0.767155528	139.7700043	8.800000191
0.8382		611.1099854	1.799999952
0.8232	0.802675188	24.86000061	44.24000168
0.7951	0.458828181	84.83000183	12.85000038
0.7064	2.07394743	472.480011	2.180000067
0.7742	0.610270143	554.9099731	1.730000019
0.7694	1.404055119	605.0999756	1.570000052
0.7612	0.75150156	138.9799957	6.260000229
0.7607		634.3300171	1.340000033
0.7461		383.1799927	2.140000105
0.7465	1.293771505	481.7099915	1.639999986
0.7747	0.577784538	63.00999832	12.38000011
0.8097	0.909752727	182.7200012	4.050000191

0.7657	0.727687597	99.29000092	7.050000191
0.7659	1.645159602	600	1.100000024
0.8082	0.761054635	590	1
0.7841	3.977871418	28.32999992	20.12000084
0.8257		550	1
0.6845	0.687594414	534.6500244	1.00999999
0.7849	0.544478476	675	0.800000012
0.8389		611.7700195	0.850000024
0.8302	0.965025246	460.7799988	1.019999981
0.859		642.8599854	0.699999988
0.7801	0.825483918	478.2600098	0.920000017
0.7465	0.407469511	27.04000092	15.89999962
0.7851	0.298336804	23.54999924	18.26000023
0.8514		641.789978	0.670000017
0.6189		640.6300049	0.639999986
0.7464	1.181138635	571.4299927	0.629999995
1		666.6699829	0.50999999
1	0.450854242	666.6699829	0.50999999
0.7679	0.307640314	57.38999939	5.75
0.7535		278.2600098	1.149999976
0.7759	0.238775149	301.8900146	1.059999943
0.8068		436.6199951	0.709999979
1		24.19000053	11.15999985
0.845		590.9099731	0.439999998
0.8689	1.448701262	650	0.400000006
0.7107	0.580171943	23.36000061	10.69999981
0.7144	0.701825082	480	0.5
0.8354	0.342866451	615.3900146	0.389999986
0.7128	0.715068996	24.42000008	9.010000229
0.7657		611.1099854	0.360000014
0.7577		448.980011	0.49000001
0.836		523.8099976	0.419999987
0.7151	1.074562073	124.1800003	1.529999971
0.7822		354.1700134	0.479999989
0.8555	0.674476624	435.8999939	0.389999986
1		653.8499756	0.25999999
1		666.6699829	0.239999995
0.7648		636.3599854	0.219999999
0.687		590.9099731	0.219999999
0.7677	0.846217513	123.7099991	0.970000029
0.7143		647.0599976	0.170000002
1		647.0599976	0.170000002
0.7443	0.416463196	54.65000153	1.830000043
0.776	0.396791399	230.7700043	0.389999986
1	1.025670409	642.8599854	0.140000001

0.8551	0.979731321	450	0.200000003
0.7029	1.066494584	600	0.150000006
0.7456	0.82254225	47.61999893	1.889999986
0.662		600	0.150000006
0.7666	0.652301788	142.8600006	0.560000002
0.7504		155.5599976	0.449999988
0.8693		636.3599854	0.109999999
0.7321		111.1100006	0.540000021
0.7937		400	0.150000006
1	0.442665249	625	0.079999998
0.6994	0.705125153	555.5599976	0.090000004
0.8624		571.4299927	0.07
0.7435		500	0.079999998
1		750	0.039999999
1		600	0.050000001
0.861		500	0.039999999
0.8551		250	0.039999999
0.6864		1000	0.01
1	1.012285233	20.82999992	0.479999989
1		1000	0.01
1		1000	0.01
0.7315	0.474347025	47.61999893	0.209999993
0.5781	0.374291241	0	0.140000001

GDP ($B)	GDP/cap ($)	Gov Efficiency	K1 ETA (years)
26966	18999	0.2376	21.1
19493.2	56432	0.2012	35
10476.2	7220	0.2156	42.2
3731.4	25766	0.2393	31.5
4774.5	38581	0.2993	5.9
2137.6	41331	0.2257	32.7
1572.2	17170	0.2093	18
3500.1	12347	0.1728	24.2
1247.9	53755	0.2702	29.9
723.6	11304	0.1549	82.8
2033.4	15539	0.2264	19.9
2386	27211	0.1139	273.5
3909.6	46501	0.3137	50.8
1344.3	50322	0.0327	1750.3
833.8	8256	0.1363	87.1
1460.4	12532	0.0849	335.4
877.6	24681	0.1158	617
1761.3	44318	0.275	26.9
1124.1	15685	0.1384	230.5
527.6	11706	0.1392	274.3
1231	32276	0.1397	323
503.6	24457	0.2013	18.7
393.6	11039	0.1435	90.9
3187.4	15035	0.2603	4.8
858.1	4944	0.1352	128.1
1014.8	8760	0.1304	143.9
2136	36114	0.1101	238.9
1318.8	5249	0.2494	14.1
2615.2	37601	0.1983	25.3
595.8	12907	0.1736	51.4
219	44373	0.1388	183.9
854.9	18709	0.2544	10.8
1625.1	33935	0.1918	14.9
97.5	13240	0.0342	229
336.7	36378	0.126	25.8
381.8	125238	0.0781	1046.6
879.6	47943	0.2408	31.7
58.8	37441	0.1212	92.6
343	32327	0.2354	36.8
91.4	12517	0.1111	1442.3
309.2	7532	0.2021	11.8
116.8	17467	0.1694	73.2
356.7	47925	0.164	213.7

311.7	8184	0.1554	43.8
742.7	14044	0.1385	44.3
2671.6	40083	0.2609	17.8
448.3	22684	0.1111	274
177.4	33580	0.1788	161.3
1134.4	4876	0.1529	1679.4
263.9	26552	0.1366	369.1
175.4	16966	0.1814	35.5
341.7	9985	0.1606	343.1
149.7	5289	0.1512	130.3
175.8	19408	0.2122	5.8
191.2	16730	0.1464	350.1
291	5340	0.1544	192.6
52.4	2331	0.1625	127.7
504.7	26694	0.2053	30.9
113.9	35129	0.18	125.1
99.3	8820	0.1434	49.5
55	7169	0.1363	24.7
129.5	10545	0.1608	259.3
145.4	4304	0.1298	77.7
489.7	41644	0.1902	109.9
138.4	20610	0.23	18.2
84.2	7637	0.1213	896.1
40.5	1538	0.1458	162.9
398.8	43761	0.1175	107.4
308.4	58099	0.1378	35.3
65.1	3693	0.1294	162.4
175	9651	0.1602	212.4
44.8	12893	0.1542	527.8
285.4	29628	0.1887	29.9
295.7	28402	0.0851	110.9
37.5	24960	0.1593	101.1
494.2	46376	0.1631	39.8
78.2	6297	0.1553	75.3
200.6	38474	0.1504	324.5
		0.1561	123.2
482.3	85777	0.1666	53.6
226.2	40217	0.1932	6.4
121.5	3898	0.1742	84.2
51.9	2872	0.181	18.7
198.2	2975	0.1357	194.7
299.4	49884	0.1406	190.3
23.7	3830	0.1322	49.2
146.2	8068	0.1623	20.2
		0.1278	648

49.7	4673	0.1421	28.7
94.8	21253	0.1674	20.7
23.8	7851	0.1631	21.8
28.8	15862	0.1633	346.7
158.3	4309	0.1502	35.2
34.3	24990	0.0989	352.2
31.7	4412	0.1337	287.4
150.4	27464	0.2211	133.2
69.4	32774	0.2384	62.3
20	7051	0.1583	48.2
31.4	935	0.175	36
25.9	1585	0.1564	47.3
		0.1627	128.3
82.7	2916	0.1374	143.2
102.3	26580	0.1155	109.2
559.4	63623	0.1291	447.9
39.3	15843	0.1756	15.8
39.7	13357	0.1174	399
68.5	3304	0.1771	14.7
50.5	4772	0.2596	92.1
27	21191	0.1739	59.9
70.5	18523	0.1582	41.5
40	29757	0.1401	40.4
60.9	1545	0.0813	250.1
		0.1568	152.5
30.6	1285	0.1727	443.8
48.7	8442	0.206	73.2
69.1	20394	0.1227	18.3
36	14502	0.0806	1028.9
		0.0371	277.9
38.6	1268	0.1411	172.1
32.8	4802	0.1584	137.3
84.8	29968	0.1374	358.1
		0.0676	134.3
65.1	9516	0.0826	34
190.4	3373	0.1271	469.2
18.3	33544	0.1839	52.5
37.3	1621	0.1699	78.5
20.2	10929	0.154	878.6
51.2	27649	0.155	21.8
		0.1868	34.2
		0.1528	73
15.7	3215	0.1817	60.5
80.3	15662	0.1571	27.5
23.7	1644	0.1235	66.5

59.3	9351	0.1597	112.2
3.4	12070	0.1739	193
33.3	2359	0.0888	74.9
16	41334	0.1089	20.4
		0.0861	109.5
17.6	1526	0.1946	278.1
24.8	947	0.1567	40.5
		0.0698	96.1
24.1	4538	0.0705	721.8
		0.0465	245.1
14.3	1533	0.1406	72.3
91.3	863	0.188	11
270	2098	0.1343	18.6
		0.0644	101.7
		0.2747	4464.7
3.1	23862	0.1172	19
		0	
4.7	1726	0	
102	2156	0.1424	43.1
		0.158	41.9
28.5	2040	0.1399	72.8
		0.1333	1986.2
		0	
		0.0557	80.4
1.7	9783	0.0397	144.3
96	3229	0.2052	24.3
4.3	8179	0.1983	599.2
24.1	1307	0.071	46.2
36.2	12865	0.159	194.9
		0.0806	29.8
		0.1577	26
		0.0542	169.9
36.1	53094	0.1656	299.4
		0.155	64.2
4.2	770	0.0491	179
		0	
		0	
		0.0751	68.2
		0.0652	74.8
53.3	1286	0.1668	281.5
		0.0593	98.6
		0	
26.5	1287	0.1592	38.3
9.2	670	0.1009	81.7
1.7	1978	0	

3.4	2945	0.0255	645.9
0.7	10147	0.1788	552.5
21.9	7377	0.1681	8.5
		0.2023	1189.4
9.6	7798	0.1631	31.7
		0.1625	32.6
		0.023	752
		0.1649	373.3
		0.1127	46.9
3.3	1548	0	
0.8	3646	0.1632	37.5
		0.0359	240.7
		0.0961	19.7
		0	
		0	
		0.0249	681.3
		0.0454	325.8
		0.0674	217
4.8	2083	0	
		0	
		0	
11.7	1381	0.1639	84.6
3.8	736	0.1799	161.5

Country	GDP/cap ($)	Population (M)	Fossil Share %	K1 Geodesic
Brazil	15035	212	10.55087757	2.9029
Belarus	19408	9.1	62.50820923	6.4394
Namibia	7377	3	2.116402149	21.485
Argentina	18709	45.7	58.76336288	4.5558
Democratic Republic of Cor	863	105.8	0	20.3762
Ukraine	7532	41	27.82710075	5.0082
Pakistan	5249	251.3	53.64085388	5.1181
Zambia	3304	20.7	10.99691772	21.269
Botswana	15843	2.5	99.61389923	23.5566
Iran	17170	91.6	92.82447815	9.2505
Ethiopia	2098	128.7	0	18.5748
Senegal	2872	18.1	78.26087189	16.3043
Mexico	15539	130.9	75.41809845	5.01
Guatemala	8068	18.1	25.48291206	20.2472
China	18999	1419.3	61.83478165	4.2067
Moldova	7851	3	88.29174042	19.3156
Indonesia	12347	283.5	81.91042328	16.8468
Nepal	3229	29.7	0	18.6332
Laos	7169	7.7	23.28871918	21.3589
Costa Rica	15662	5.1	5.97738266	20.7463
Honduras	4673	10.6	38.42869186	22.0622
Eswatini	7798	1.2	3.571428537	22.8979
Paraguay	9516	6.8	0	19.0953
Angola	4309	36.7	23.57859421	22.5884
Azerbaijan	16966	10.3	88.04425049	18.4899
Mozambique	935	33.6	16.30879402	21.6537
Sao Tome and Principe	3646	0.2	88.8888855	22.6343
Malawi	1287	20.6	4.371584415	22.2418
Niger	947	26.2	97.49999237	25.7602
Georgia	18523	3.8	19.8457222	22.0245
India	7220	1450.9	77.74752045	5.0513
Uganda	2156	47.3	2.608695745	22.6965
Morocco	8184	38.1	73.49589539	16.6347
Colombia	14044	52.9	35.64163971	21.6281
Chad	1307	18.5	94.87180328	22.9289
Zimbabwe	1585	16.3	32.49097443	22.2484
Jamaica	7051	2.8	87.08241272	19.8623
Congo	3830	6.2	79.30367279	22.272
Jordan	8820	11.3	76.87195587	21.9939
Algeria	12907	46.2	99.05503082	19.8041
Mauritania	3215	4.9	72.56098175	23.1482
Guinea	1644	14.4	25.18518448	22.9891
Togo	1533	9.3	79.3478241	24.4849

Rwanda	2040	14	43.39622879	24.4544
Lebanon	8442	5.8	52.65486908	21.2673
Serbia	17467	6.7	72.97953033	16.2651
Benin	2359	14.1	97	24.5843
Bolivia	6297	12.4	62.02428818	19.4466
Ghana	4304	33.8	61.46562195	22.0199
Burkina Faso	1621	23	82.65895844	23.3945
Burundi	670	13.7	30.7692318	23.9249
South Africa	11304	64	83.81177521	14.2403
Cote d'Ivoire	3898	31.2	68.91284943	21.5391
Sierra Leone	1381	8.5	4.761904716	23.0133
Vietnam	8256	101	57.71790314	16.1346
Uzbekistan	11039	35.7	90.75409698	23.6639
Tajikistan	4772	10.6	8.579336166	23.4499
El Salvador	9351	6.3	6.808510303	21.7365
Syria	2331	22.5	95.61753082	22.5822
Bangladesh	4944	173.6	97.97119141	24.2212
Venezuela	5289	28.3	21.61803627	21.5492
Nicaragua	4802	6.8	35.46910858	21.9424
Cameroon	2916	28.4	36.13445282	22.615
Philippines	8760	115.8	78.26950836	16.4138
Saint Lucia	9783	0.2	97.49999237	21.6499
Central African Republic	736	5.1	0	16.8166
Cambodia	3693	17.6	59.28752899	22.9426
North Korea	1538	26.3	36.91149139	23.9564
Madagascar	1268	30.4	64.77272797	23.5473
Liberia	770	5.5	66.66666412	22.9995
Myanmar	5340	54.5	59.76499176	21.0871
Barbados	12070	0.3	91.81817627	24.1298
Tanzania	2975	66.6	74.4990921	21.7801
Albania	12865	2.8	0	19.3472
Ecuador	9651	18.1	28.33333397	19.296
Turkmenistan	13240	7.4	99.97039795	19.324
Thailand	15685	71.7	85.15938568	16.1381
Yemen	1545	39.4	83.06188965	25.0118
Tunisia	10545	12.3	96.0153656	20.8516
Iraq	11706	45.1	98.82893372	23.7278
Haiti	1526	11.5	81.18811798	22.8363
Afghanistan	1286	41.5	13.40206051	24.328
Kyrgyzstan	4412	7.2	17.43684578	25.0368
Egypt	12532	116.5	88.80323792	22.2652
Peru	9985	34.2	40.72043991	15.7724
North Macedonia	15862	1.8	63.4974556	16.449
Dominican Republic	16730	11.4	81.45720673	16.8634
Armenia	13357	3	39.595047	22.9229

Mali	1285	23.8	57.33944702	23.3447
Kenya	3373	56.4	8.093384743	21.7979
Mongolia	12893	3.5	90.33412933	22.1873
Dominica	10147	0.1	86.66665649	22.7074
Cape Verde	8179	0.5	72	23.4447
Djibouti	2945	1.2	65	22.6193
Palestine	4538	5.3	66.66667175	22.6115
Equatorial Guinea	10929	1.8	68.78981018	24.104
Cuba	7637	11	95.29103851	19.4293
Gabon	14502	2.5	52.03761673	24.7247
Libya	12517	7.3	99.97151184	23.5538
Nigeria	4876	232.7	76.97483063	21.3218
Comoros	1978	0.9	100	19.5593
Gambia	1726	2.7	100	19.5593
Guinea-Bissau	1548	2.1	100	19.5593
Lesotho	2083	2.3	0	16.8166

K1 ETA (years)	χ̄	CV(χ)	Gov Efficiency	Stagnation Cost ($M/yr)
4.8	0.467487	0.7962	0.2603	3954.6
5.8	0.373137	0.7588	0.2122	738.5
8.5	0.254446	0.5135	0.1681	4.5
10.8	0.482249	0.8956	0.2544	2629.5
11	0.253498	0.3485	0.188	21.5
11.8	0.392424	0.9414	0.2021	1396.5
14.1	0.423321	0.6976	0.2494	3626
14.7	0.234839	0.3257	0.1771	108
15.8	0.252584	0.4385	0.1756	110
18	0.378956	0.8102	0.2093	12837.1
18.6	0.214851	0.5999	0.1343	21.5
18.7	0.29293	0.6187	0.181	215.5
19.9	0.36743	0.6226	0.2264	8580.1
20.2	0.288322	0.7768	0.1623	183.5
21.1	0.350969	0.477	0.2376	280112.7
21.8	0.254047	0.5575	0.1631	164
24.2	0.3337	0.9307	0.1728	12637
24.3	0.289348	0.4098	0.2052	12.5
24.7	0.262722	0.9282	0.1363	607
27.5	0.225339	0.434	0.1571	39
28.7	0.237952	0.6745	0.1421	169.5
31.7	0.233361	0.4308	0.1631	4
34	0.176806	1.1398	0.0826	55
35.2	0.219026	0.4582	0.1502	150
35.5	0.276258	0.5228	0.1814	886.5
36	0.245145	0.4007	0.175	125
37.5	0.300591	0.8414	0.1632	2.5
38.3	0.255737	0.6061	0.1592	5
40.5	0.215101	0.3723	0.1567	27
41.5	0.224895	0.422	0.1582	102
42.2	0.378048	0.7538	0.2156	71813.2
43.1	0.232107	0.6302	0.1424	16.5
43.8	0.292601	0.8829	0.1554	1265
44.3	0.256606	0.8532	0.1385	1250.5
46.2	0.164619	1.3198	0.071	12
47.3	0.224702	0.4366	0.1564	124
48.2	0.261248	0.65	0.1583	126
49.2	0.223632	0.6915	0.1322	184.5
49.5	0.229132	0.5974	0.1434	608.5
51.4	0.273504	0.5757	0.1736	3051
60.5	0.253454	0.3947	0.1817	39.5
66.5	0.190275	0.5408	0.1235	37
72.3	0.219935	0.5638	0.1406	22

72.8	0.224136	0.6026	0.1399	16
73.2	0.280742	0.3631	0.206	83.5
73.2	0.298783	0.7639	0.1694	1292
74.9	0.191792	1.1603	0.0888	29.5
75.3	0.266015	0.7133	0.1553	289
77.7	0.213213	0.6427	0.1298	550
78.5	0.225844	0.3289	0.1699	48
81.7	0.223981	1.2199	0.1009	4.5
82.8	0.297025	0.9172	0.1549	8881.5
84.2	0.235096	0.3495	0.1742	219
84.6	0.268523	0.6386	0.1639	0.5
87.1	0.252593	0.8535	0.1363	7351.5
90.9	0.222779	0.5522	0.1435	4274.5
92.1	0.302466	0.1652	0.2596	106
112.2	0.234325	0.4673	0.1597	35
127.7	0.240953	0.4828	0.1625	685
128.1	0.229598	0.6982	0.1352	3950
130.3	0.242952	0.6066	0.1512	747.5
137.3	0.253153	0.5985	0.1584	63
143.2	0.232225	0.69	0.1374	119
143.9	0.277688	1.1295	0.1304	3853.5
144.3	0.131132	2.3061	0.0397	13
161.5	0.421925	1.3448	0.1799	
162.4	0.210231	0.6249	0.1294	391.5
162.9	0.219566	0.5062	0.1458	457
172.1	0.235056	0.6657	0.1411	63
179	0.144546	1.941	0.0491	8.5
192.6	0.253853	0.6439	0.1544	703
193	0.234103	0.3464	0.1739	33
194.7	0.217559	0.6027	0.1357	204
194.9	0.287248	0.8064	0.159	11
212.4	0.24266	0.5152	0.1602	346
229	0.129434	2.7881	0.0342	2207
230.5	0.271213	0.96	0.1384	5534
250.1	0.163512	1.0105	0.0813	90
259.3	0.242968	0.5114	0.1608	583.5
274.3	0.225773	0.6223	0.1392	5180.5
278.1	0.315485	0.6216	0.1946	27
281.5	0.232313	0.393	0.1668	6
287.4	0.211789	0.5835	0.1337	140
335.4	0.159076	0.8747	0.0849	6819
343.1	0.310687	0.9347	0.1606	840.5
346.7	0.337109	1.0644	0.1633	156.5
350.1	0.282267	0.9286	0.1464	734
399	0.196215	0.6718	0.1174	108

443.8	0.231255	0.3393	0.1727	86
469.2	0.20494	0.6122	0.1271	54.5
527.8	0.233982	0.5177	0.1542	338.5
552.5	0.29712	0.6622	0.1788	4.5
599.2	0.285568	0.4403	0.1983	12
645.9	0.144856	4.6904	0.0255	4.5
721.8	0.169832	1.4078	0.0705	23.5
878.6	0.230616	0.4978	0.154	47.5
896.1	0.227254	0.8736	0.1213	488.5
1028.9	0.124413	0.5434	0.0806	68.5
1442.3	0.178013	0.6019	0.1111	1458
1679.4	0.219846	0.4382	0.1529	1019
	0	0	0	4.5
	0	0	0	17
	0	0	0	2.5
	0	0	0	0.5

Fossil Trend (pp/yr)	Development Stage	Current Mix: Coal	Gas	Oil
-0.92	Post-fossil (K1 approaching)	0.0217	0.0654	0.0184
-5.82	Early transition	0.0007	0.6225	0.002
0.12	Post-fossil (K1 approaching)	0.0159	0	0.0053
-1.71	Mid-transition	0.0086	0.5373	0.0417
-0.02	Post-fossil (K1 approaching)	0	0	0
-1.73	Mid-transition	0.2091	0.0644	0.0048
-1.7	Mid-transition	0.1395	0.2496	0.1473
-0.73	Post-fossil (K1 approaching)	0.1095	0	0.0005
-0.01	Early transition	0.9575	0	0.0386
0.97	Degrading (moving away fro	0.0018	0.8592	0.0672
	Post-fossil (K1 approaching)	0	0	0
-1.26	Early transition	0.0584	0.0025	0.7217
-0.77	Early transition	0.0349	0.6241	0.0952
-1.4	Early transition	0.1806	0	0.0808
-1.11	Early transition	0.5777	0.0318	0.0088
-0.87	Early transition	0	0.8253	0.0576
-0.3	Early transition	0.6439	0.1939	0.0201
	Post-fossil (K1 approaching)	0	0	0
-1.94	Early transition	0.2329	0	0
0.85	Post-fossil (K1 approaching)	0	0	0.0684
0.43	Early transition	0	0	0.3961
-0.09	Post-fossil (K1 approaching)	0.0357	0	0
	Post-fossil (K1 approaching)	0	0	0
0.51	Early transition	0	0.0931	0.1427
-0.84	Early transition	0	0.8765	0.0039
-0.32	Post-fossil (K1 approaching)	0	0.1564	0.0066
-1.85	Early transition	0	0	0.8889
0.73	Post-fossil (K1 approaching)	0	0	0.0437
-0.1	Early transition	0.225	0.0625	0.6875
-0.68	Post-fossil (K1 approaching)	0	0.1985	0
-0.22	Early transition	0.7477	0.0278	0.002
0.13	Post-fossil (K1 approaching)	0	0	0.0261
-1.12	Early transition	0.5928	0.101	0.0412
0.85	Early transition	0.1515	0.1683	0.0366
0.19	Pre-transition (fossil locked	0	0	0.9487
-1.97	Early transition	0.3249	0	0
-0.21	Early transition	0	0.5323	0.3385
2.29	Degrading (moving away fro	0	0.7369	0.0561
-2.73	Early transition	0	0.7306	0.0381
0.02	Early transition	0	0.9865	0.004
-0.5	Early transition	0	0	0.7256
-2.07	Early transition	0	0	0.2519
2.39	Degrading (moving away fro	0	0.7174	0.0761

-2.52	Early transition	0.0472	0.2075	0.1792
-7.52	Early transition	0	0	0.5265
0.05	Early transition	0.6476	0.0808	0.0013
-0.5	Pre-transition (fossil locked	0	0.73	0.24
-0.3	Early transition	0	0.6032	0.017
0.36	Early transition	0	0.597	0.0177
-0.48	Early transition	0	0	0.8266
0.37	Early transition	0	0	0.3077
-0.99	Early transition	0.8303	0	0.0078
-0.2	Early transition	0	0.6873	0.0018
-4.76	Post-fossil (K1 approaching)	0	0	0.0476
-1.98	Early transition	0.5032	0.0719	0.002
0.02	Early transition	0.0748	0.8131	0.0197
0.01	Post-fossil (K1 approaching)	0.0706	0.0152	0
-3.69	Post-fossil (K1 approaching)	0	0	0.0878
-0.03	Early transition	0	0.3835	0.5727
-0.11	Early transition	0.2016	0.6154	0.1627
-3.34	Early transition	0	0.1463	0.0699
-0.97	Early transition	0	0	0.4212
-0.83	Early transition	0	0.2353	0.1261
-0.16	Early transition	0.6807	0.1536	0.018
-0.42	Pre-transition (fossil locked	0	0	0.975
	Post-fossil (K1 approaching)	0	0	0
1.44	Early transition	0.5611	0	0.0318
-2.1	Early transition	0.3476	0	0.0215
1.12	Degrading (moving away fro	0.1932	0	0.4545
-0.27	Unstable (burst-driven, no	0	0	0.6667
2.67	Early transition	0.0948	0.4984	0.0045
-0.88	Early transition	0	0.0273	0.8909
1.04	Degrading (moving away fro	0	0.7395	0.0055
	Post-fossil (K1 approaching)	0	0	0
1.02	Early transition	0	0.0306	0.2527
0	Pre-transition (fossil locked	0	0.9997	0
0	Early transition	0.1667	0.6835	0.0014
-0.53	Pre-transition (fossil locked	0	0.1792	0.6515
-0.14	Early transition	0	0.9501	0.0101
0.12	Early transition	0	0.5817	0.4066
-1.09	Early transition	0	0	0.8119
-0.62	Post-fossil (K1 approaching)	0.0309	0	0.1031
1.5	Post-fossil (K1 approaching)	0.1411	0.0129	0.0203
-0.24	Pre-transition (fossil locked	0	0.8129	0.0751
0.21	Mid-transition	0.0013	0.3951	0.0108
-1.46	Early transition	0.3871	0.2309	0.017
-1	Early transition	0.3011	0.32	0.1935
-0.19	Early transition	0	0.396	0

-0.42	Early transition	0	0	0.5734
-0.54	Post-fossil (K1 approaching)	0	0	0.1427
0.05	Early transition	0.8616	0	0.0418
0.81	Degrading (moving away fro	0	0	0.8667
-1.57	Early transition	0	0	0.72
-5.83	Unstable (burst-driven, no c	0	0	0.65
-4.04	Early transition	0	0	0.6667
0.23	Early transition	0	0.6815	0.0064
-0.21	Early transition	0	0.1321	0.8208
-0.14	Early transition	0	0.3511	0.1693
0	Pre-transition (fossil locked	0	0.7605	0.2392
0.05	Early transition	0	0.7697	0
	Pre-transition (fossil locked	0	0	1
	Pre-transition (fossil locked	0	0	1
	Pre-transition (fossil locked	0	0	1
	Post-fossil (K1 approaching)	0	0	0

Nuclear	Hydro	Solar	Wind	Biofuel
0.0212	0.5541	0.0956	0.1455	0.0781
0.3432	0.0107	0.007	0.0042	0.0098
0	0.6984	0.2698	0.0106	0
0.0685	0.1945	0.026	0.106	0.0172
0	0.8597	0.1384	0	0.0019
0.5576	0.0988	0.0456	0.0127	0.007
0.1345	0.1887	0.0997	0.0299	0.0107
0	0.8782	0.0077	0	0.0041
0	0	0.0039	0	0
0.0186	0.0477	0.0023	0.0032	0.0001
0	0.9665	0.0022	0.0307	0.0005
0	0.0385	0.0733	0.0919	0.0137
0.0347	0.0663	0.0724	0.0556	0.0169
0	0.4306	0.0183	0.0259	0.2637
0.0447	0.1343	0.0832	0.0988	0.0207
0	0.0461	0.0288	0.0403	0.0019
0	0.0769	0.0038	0.0013	0.0601
0	0.9897	0.0093	0.0009	0
0	0.7648	0.0015	0	0.0008
0	0.8022	0.0074	0.1165	0.0055
0	0.3415	0.0915	0.0607	0.11
0	0.5357	0.0536	0	0.375
0	0.9946	0	0	0.0054
0	0.7397	0.0217	0	0.0028
0	0.1021	0.0107	0.0029	0.0039
0	0.8267	0.0036	0	0.0066
0	0.1111	0	0	0
0	0.9235	0.0055	0	0.0273
0	0	0.025	0	0
0	0.7952	0	0.0063	0
0.0269	0.0771	0.0674	0.0402	0.011
0	0.8661	0.0261	0	0.0817
0	0.0153	0.0362	0.2126	0.0009
0	0.581	0.0367	0	0.0259
0	0	0	0.0256	0.0256
0	0.657	0.0036	0	0.0144
0	0.0267	0.029	0.0601	0.0134
0	0.2031	0.0019	0	0.0019
0	0.0009	0.1529	0.0775	0
0	0.0002	0.009	0.0002	0
0	0.128	0.0854	0.061	0
0	0.7407	0.0074	0	0
0	0.087	0.1196	0	0

0	0.5283	0.0377	0	0
0	0.1549	0.3097	0.0022	0.0066
0	0.2247	0.0013	0.0353	0.0089
0	0	0.03	0	0
0	0.2834	0.0275	0.0356	0.0332
0	0.3783	0.0062	0	0.0008
0	0.0694	0.052	0	0.052
0	0.6667	0	0	0.0256
0.0313	0.0044	0.0797	0.0446	0.0019
0	0.301	0.0018	0	0.0081
0	0.8571	0.0952	0	0
0	0.2922	0.0852	0.042	0.0034
0	0.0866	0.0059	0	0
0	0.9142	0	0	0
0	0.4845	0.2358	0.0201	0.1718
0	0.0374	0.005	0	0.0015
0	0.0078	0.0125	0	0
0	0.7835	0.0001	0.0002	0
0	0.1766	0.0082	0.1522	0.2418
0	0.6315	0.0024	0	0.0048
0	0.094	0.0318	0.0106	0.0112
0	0	0.025	0	0
0	1	0	0	0
0	0.3372	0.0681	0	0.0019
0	0.6252	0.0056	0	0
0	0.3106	0.0303	0	0.0114
0	0.3333	0	0	0
0	0.3861	0.0053	0	0.0109
0	0	0.0818	0	0
0	0.2459	0.0027	0	0.0064
0	0.9667	0.0333	0	0
0	0.6973	0.0012	0.0024	0.0158
0	0.0003	0	0	0
0	0.0321	0.0272	0.0169	0.0721
0	0	0.1694	0	0
0	0.0005	0.0235	0.0158	0
0	0.0092	0.0025	0	0
0	0.1881	0	0	0
0	0.7732	0.0928	0	0
0	0.8256	0	0	0
0	0.0605	0.0256	0.0259	0
0	0.5018	0.0202	0.0611	0.0097
0	0.2258	0.0917	0.0374	0.0102
0	0.0694	0.0652	0.0424	0.0085
0.3048	0.2036	0.0956	0	0

0	0.3761	0.0344	0	0.0161
0	0.4952	0.0754	0.2483	0.0384
0	0.0072	0.0227	0.0668	0
0	0.1333	0	0	0
0	0	0.14	0.14	0
0	0	0	0.35	0
0	0	0.3333	0	0
0	0.3121	0	0	0
0	0.0078	0.0137	0.0039	0.0216
0	0.4765	0	0	0.0031
0	0	0.0003	0	0
0	0.2263	0.0025	0	0.0015
0	0	0	0	0
0	0	0	0	0
0	0	0	0	0
0	1	0	0	0

Geodesic Path: Australia → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.4963	0.4521	0.1743	0.0228	0	0.0456
1	0.05	1.5487	0.4156	0.1715	0.0235	0	0.052
2	0.1	1.5948	0.3792	0.1675	0.024	0	0.0589
3	0.15	1.634	0.3434	0.1623	0.0244	0	0.0661
4	0.2	1.6658	0.3086	0.1561	0.0246	0	0.0737
5	0.25	1.6901	0.2752	0.149	0.0246	0	0.0815
6	0.3	1.7072	0.2436	0.1412	0.0244	0	0.0895
7	0.35	1.7173	0.2141	0.1328	0.0241	0	0.0976
8	0.4	1.7212	0.1869	0.1241	0.0236	0	0.1057
9	0.45	1.7198	0.162	0.1151	0.0229	0	0.1137
10	0.5	1.714	0.1396	0.1062	0.0222	0.0001	0.1215
11	0.55	1.7049	0.1196	0.0974	0.0213	0.0001	0.1291
12	0.6	1.6938	0.1019	0.0888	0.0204	0.0003	0.1364
13	0.65	1.6819	0.0864	0.0805	0.0193	0.0008	0.1434
14	0.7	1.6713	0.0728	0.0727	0.0183	0.0017	0.15
15	0.75	1.6644	0.0611	0.0652	0.0172	0.0039	0.156
16	0.8	1.665	0.0509	0.0582	0.0161	0.0087	0.1612
17	0.85	1.6776	0.042	0.0514	0.0149	0.0193	0.165
18	0.9	1.7057	0.0341	0.0446	0.0136	0.0423	0.1662
19	0.95	1.7434	0.0269	0.0376	0.012	0.0897	0.1624
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Brazil → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.4478	0.0217	0.0654	0.0184	0.0212	0.5541
1	0.05	1.4863	0.0222	0.0646	0.0183	0.0242	0.5328
2	0.1	1.5233	0.0227	0.0636	0.0182	0.0275	0.511
3	0.15	1.5586	0.023	0.0624	0.018	0.0313	0.4889
4	0.2	1.592	0.0234	0.0612	0.0178	0.0355	0.4666
5	0.25	1.6232	0.0237	0.0598	0.0175	0.0402	0.4441
6	0.3	1.652	0.0239	0.0583	0.0172	0.0453	0.4215
7	0.35	1.6784	0.024	0.0566	0.0169	0.0509	0.399
8	0.4	1.702	0.0241	0.0549	0.0165	0.0571	0.3766
9	0.45	1.7228	0.0241	0.053	0.0161	0.0639	0.3544
10	0.5	1.7406	0.0241	0.0511	0.0156	0.0712	0.3326
11	0.55	1.7555	0.0239	0.0491	0.0152	0.0792	0.3112
12	0.6	1.7674	0.0237	0.047	0.0146	0.0877	0.2904
13	0.65	1.7762	0.0235	0.0449	0.0141	0.097	0.2701
14	0.7	1.7821	0.0231	0.0428	0.0135	0.1069	0.2505
15	0.75	1.785	0.0227	0.0406	0.013	0.1174	0.2316
16	0.8	1.785	0.0223	0.0384	0.0124	0.1286	0.2136
17	0.85	1.7823	0.0218	0.0363	0.0118	0.1405	0.1963
18	0.9	1.7769	0.0212	0.0341	0.0112	0.153	0.18

19	0.95	1.769	0.0206	0.0321	0.0106	0.1662	0.1645
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: China → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.3926	0.5777	0.0318	0.0088	0.0447	0.1343
1	0.05	1.4904	0.5275	0.0342	0.0096	0.0518	0.1458
2	0.1	1.5794	0.4765	0.0365	0.0103	0.0593	0.1567
3	0.15	1.6578	0.4258	0.0385	0.011	0.0672	0.1666
4	0.2	1.7241	0.3765	0.0401	0.0115	0.0754	0.1753
5	0.25	1.7778	0.3294	0.0414	0.012	0.0837	0.1824
6	0.3	1.8189	0.2854	0.0423	0.0124	0.092	0.188
7	0.35	1.8482	0.2449	0.0429	0.0127	0.1001	0.192
8	0.4	1.867	0.2084	0.043	0.0128	0.1081	0.1944
9	0.45	1.8767	0.176	0.0429	0.0129	0.1158	0.1953
10	0.5	1.879	0.1476	0.0424	0.0129	0.1231	0.1948
11	0.55	1.8755	0.123	0.0417	0.0128	0.1302	0.1931
12	0.6	1.8676	0.1019	0.0408	0.0126	0.1368	0.1905
13	0.65	1.8566	0.0841	0.0397	0.0124	0.1432	0.1869
14	0.7	1.8437	0.0691	0.0384	0.0121	0.1492	0.1827
15	0.75	1.8296	0.0565	0.0371	0.0118	0.155	0.1779
16	0.8	1.815	0.0461	0.0357	0.0115	0.1604	0.1728
17	0.85	1.8003	0.0376	0.0343	0.0111	0.1656	0.1673
18	0.9	1.7859	0.0305	0.0329	0.0108	0.1706	0.1616
19	0.95	1.772	0.0247	0.0314	0.0104	0.1754	0.1558
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: France → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.1253	0.0031	0.0304	0.0182	0.6906	0.1046
1	0.05	1.1808	0.0035	0.0314	0.0183	0.6674	0.1101
2	0.1	1.2362	0.0039	0.0323	0.0183	0.643	0.1155
3	0.15	1.291	0.0044	0.0332	0.0182	0.6177	0.1209
4	0.2	1.3447	0.005	0.034	0.0181	0.5915	0.126
5	0.25	1.3968	0.0056	0.0346	0.018	0.5644	0.131
6	0.3	1.4467	0.0063	0.0352	0.0177	0.5368	0.1356
7	0.35	1.4941	0.007	0.0357	0.0174	0.5086	0.14
8	0.4	1.5384	0.0077	0.036	0.0171	0.4802	0.1439
9	0.45	1.5793	0.0085	0.0362	0.0167	0.4516	0.1474
10	0.5	1.6164	0.0094	0.0362	0.0162	0.4231	0.1504
11	0.55	1.6494	0.0103	0.0361	0.0157	0.3949	0.1528
12	0.6	1.6781	0.0113	0.0359	0.0152	0.3672	0.1547
13	0.65	1.7025	0.0122	0.0356	0.0146	0.3401	0.1561
14	0.7	1.7225	0.0133	0.0351	0.014	0.3138	0.1568
15	0.75	1.7381	0.0143	0.0345	0.0133	0.2885	0.157

16	0.8	1.7496	0.0154	0.0337	0.0127	0.2643	0.1566
17	0.85	1.7571	0.0165	0.0329	0.012	0.2413	0.1557
18	0.9	1.7609	0.0177	0.032	0.0113	0.2195	0.1543
19	0.95	1.7613	0.0188	0.031	0.0107	0.1991	0.1523
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Germany → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.7698	0.2074	0.1662	0.038	0	0.0393
1	0.05	1.7733	0.1909	0.1579	0.0368	0	0.0435
2	0.1	1.7742	0.1752	0.1495	0.0355	0	0.048
3	0.15	1.7725	0.1604	0.1412	0.0342	0	0.0528
4	0.2	1.7686	0.1463	0.1329	0.0328	0	0.0579
5	0.25	1.7626	0.1331	0.1248	0.0314	0	0.0633
6	0.3	1.7548	0.1208	0.1168	0.0299	0	0.069
7	0.35	1.7455	0.1093	0.1091	0.0285	0	0.0751
8	0.4	1.7348	0.0986	0.1015	0.027	0	0.0814
9	0.45	1.7231	0.0888	0.0943	0.0256	0	0.0881
10	0.5	1.7107	0.0797	0.0874	0.0241	0.0001	0.0951
11	0.55	1.6979	0.0714	0.0808	0.0227	0.0001	0.1024
12	0.6	1.6851	0.0639	0.0745	0.0214	0.0003	0.11
13	0.65	1.6732	0.057	0.0686	0.02	0.0007	0.1179
14	0.7	1.6633	0.0507	0.063	0.0188	0.0015	0.1261
15	0.75	1.6575	0.045	0.0576	0.0175	0.0035	0.1344
16	0.8	1.6591	0.0397	0.0525	0.0162	0.0079	0.1427
17	0.85	1.6727	0.0348	0.0475	0.015	0.018	0.1504
18	0.9	1.702	0.0301	0.0424	0.0136	0.0403	0.1562
19	0.95	1.7418	0.0252	0.0367	0.012	0.0877	0.1575
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: India → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	0.9848	0.7477	0.0278	0.002	0.0269	0.0771
1	0.05	1.1127	0.6998	0.0313	0.0024	0.0332	0.0894
2	0.1	1.2416	0.6468	0.0348	0.0029	0.0405	0.1024
3	0.15	1.3671	0.5898	0.0382	0.0035	0.0487	0.1157
4	0.2	1.4845	0.53	0.0413	0.0041	0.0576	0.1288
5	0.25	1.5895	0.4691	0.044	0.0047	0.0672	0.1413
6	0.3	1.6789	0.4089	0.0461	0.0053	0.0772	0.1526
7	0.35	1.7508	0.3511	0.0476	0.0059	0.0873	0.1623
8	0.4	1.8047	0.2971	0.0485	0.0065	0.0974	0.1702
9	0.45	1.8416	0.2481	0.0487	0.007	0.1072	0.1761
10	0.5	1.8634	0.2046	0.0483	0.0075	0.1165	0.18
11	0.55	1.8728	0.167	0.0474	0.008	0.1253	0.182
12	0.6	1.8726	0.135	0.0461	0.0084	0.1335	0.1823

13	0.65	1.8653	0.1082	0.0445	0.0087	0.1411	0.1811
14	0.7	1.8534	0.0862	0.0426	0.009	0.1481	0.1787
15	0.75	1.8385	0.0682	0.0406	0.0092	0.1545	0.1752
16	0.8	1.8223	0.0538	0.0385	0.0095	0.1604	0.1711
17	0.85	1.8056	0.0422	0.0363	0.0096	0.1658	0.1663
18	0.9	1.7892	0.033	0.0342	0.0098	0.1709	0.1611
19	0.95	1.7735	0.0257	0.0321	0.0099	0.1756	0.1556
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Indonesia → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.0763	0.6439	0.1939	0.0201	0	0.0769
1	0.05	1.1486	0.6115	0.1995	0.0219	0	0.0899
2	0.1	1.2223	0.5769	0.204	0.0237	0	0.1043
3	0.15	1.2966	0.5403	0.207	0.0255	0	0.1201
4	0.2	1.3703	0.5018	0.2083	0.0272	0	0.1372
5	0.25	1.4424	0.4618	0.2077	0.0288	0	0.1553
6	0.3	1.5118	0.4207	0.2051	0.0302	0	0.174
7	0.35	1.5771	0.3791	0.2002	0.0312	0	0.1929
8	0.4	1.637	0.3375	0.1932	0.0319	0	0.2112
9	0.45	1.6905	0.2965	0.1839	0.0322	0	0.2282
10	0.5	1.7362	0.2569	0.1726	0.0321	0.0001	0.2432
11	0.55	1.7733	0.2192	0.1596	0.0314	0.0002	0.2552
12	0.6	1.8008	0.184	0.1452	0.0303	0.0005	0.2635
13	0.65	1.8184	0.1519	0.1299	0.0288	0.0012	0.2675
14	0.7	1.8267	0.1231	0.1141	0.0268	0.0026	0.2667
15	0.75	1.8269	0.0978	0.0982	0.0244	0.0057	0.2607
16	0.8	1.822	0.0761	0.0828	0.0218	0.0121	0.2495
17	0.85	1.8151	0.0578	0.0681	0.0191	0.0252	0.2329
18	0.9	1.8082	0.0425	0.0543	0.0161	0.0509	0.2109
19	0.95	1.7961	0.03	0.0416	0.0131	0.0986	0.1831
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Japan → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.643	0.3219	0.3407	0.0246	0.0835	0.0782
1	0.05	1.7016	0.3003	0.3234	0.0252	0.0931	0.0866
2	0.1	1.7563	0.2782	0.3049	0.0257	0.1029	0.0952
3	0.15	1.806	0.2558	0.2853	0.0259	0.113	0.1039
4	0.2	1.8496	0.2334	0.2648	0.026	0.1231	0.1126
5	0.25	1.8862	0.2112	0.2439	0.0258	0.133	0.1209
6	0.3	1.9155	0.1896	0.2228	0.0255	0.1426	0.1289
7	0.35	1.9372	0.1688	0.2019	0.0249	0.1516	0.1362
8	0.4	1.9514	0.1492	0.1815	0.0242	0.1599	0.1428
9	0.45	1.9584	0.1307	0.1619	0.0233	0.1673	0.1486

10	0.5	1.9589	0.1137	0.1433	0.0222	0.1738	0.1535
11	0.55	1.9536	0.0982	0.1259	0.0211	0.1792	0.1573
12	0.6	1.9432	0.0842	0.1099	0.0199	0.1835	0.1602
13	0.65	1.9287	0.0718	0.0953	0.0186	0.1867	0.162
14	0.7	1.9107	0.0608	0.0821	0.0173	0.1887	0.1628
15	0.75	1.89	0.0511	0.0703	0.016	0.1897	0.1627
16	0.8	1.867	0.0428	0.0599	0.0147	0.1896	0.1617
17	0.85	1.8421	0.0357	0.0508	0.0134	0.1885	0.1598
18	0.9	1.8157	0.0295	0.0428	0.0122	0.1865	0.1572
19	0.95	1.7879	0.0244	0.0359	0.0111	0.1836	0.1539
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Nigeria → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	0.5623	0	0.7697	0	0	0.2263
1	0.05	0.6011	0	0.7428	0	0	0.2516
2	0.1	0.6408	0	0.7135	0	0	0.2785
3	0.15	0.6815	0	0.682	0	0	0.3067
4	0.2	0.7232	0	0.6483	0	0	0.3359
5	0.25	0.7664	0	0.6124	0	0	0.3656
6	0.3	0.8112	0	0.5745	0	0	0.3952
7	0.35	0.8581	0	0.5347	0	0	0.4237
8	0.4	0.9073	0	0.4931	0	0	0.4503
9	0.45	0.9592	0	0.4502	0	0	0.4737
10	0.5	1.0139	0	0.4062	0	0.0001	0.4925
11	0.55	1.0713	0.0001	0.3616	0.0001	0.0003	0.5052
12	0.6	1.1313	0.0002	0.317	0.0001	0.0007	0.5102
13	0.65	1.1946	0.0004	0.2729	0.0002	0.0015	0.5061
14	0.7	1.2628	0.0007	0.23	0.0005	0.0035	0.4915
15	0.75	1.3404	0.0015	0.1889	0.0009	0.0077	0.4651
16	0.8	1.4332	0.0029	0.15	0.0016	0.0166	0.4256
17	0.85	1.545	0.0053	0.1138	0.0029	0.0342	0.3719
18	0.9	1.6654	0.0091	0.0808	0.0049	0.0659	0.3043
19	0.95	1.7557	0.0143	0.0523	0.0074	0.1156	0.2268
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Poland → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.4613	0.5078	0.1443	0.038	0	0.0099
1	0.05	1.5254	0.468	0.1446	0.0385	0	0.0123
2	0.1	1.5829	0.4278	0.1436	0.0387	0	0.0151
3	0.15	1.6327	0.3878	0.1415	0.0386	0	0.0184
4	0.2	1.6739	0.3486	0.1382	0.0381	0	0.0223
5	0.25	1.7063	0.3106	0.1338	0.0374	0	0.0268
6	0.3	1.7297	0.2745	0.1285	0.0363	0	0.0319

7	0.35	1.7446	0.2405	0.1224	0.035	0	0.0376
8	0.4	1.7517	0.2091	0.1156	0.0334	0	0.0441
9	0.45	1.7519	0.1805	0.1084	0.0317	0	0.0512
10	0.5	1.7463	0.1546	0.1009	0.0299	0.0001	0.0591
11	0.55	1.7364	0.1315	0.0933	0.028	0.0002	0.0677
12	0.6	1.7235	0.1112	0.0858	0.026	0.0003	0.0771
13	0.65	1.7093	0.0935	0.0783	0.024	0.0008	0.0872
14	0.7	1.6958	0.0781	0.0711	0.0221	0.0018	0.0982
15	0.75	1.6857	0.0648	0.0642	0.0202	0.004	0.1098
16	0.8	1.6828	0.0534	0.0575	0.0183	0.0089	0.1218
17	0.85	1.6916	0.0436	0.051	0.0164	0.0197	0.1339
18	0.9	1.7154	0.035	0.0444	0.0145	0.0429	0.1446
19	0.95	1.7482	0.0272	0.0375	0.0124	0.0903	0.1515
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: South Africa → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	0.6767	0.8303	0	0.0078	0.0313	0.0044
1	0.05	0.7889	0.7905	0	0.009	0.0392	0.006
2	0.1	0.9086	0.7441	0	0.0104	0.0485	0.0082
3	0.15	1.0322	0.6912	0	0.0118	0.0593	0.0109
4	0.2	1.1553	0.6326	0	0.0131	0.0713	0.0143
5	0.25	1.2729	0.5695	0	0.0144	0.0844	0.0186
6	0.3	1.3799	0.5037	0	0.0156	0.0982	0.0236
7	0.35	1.4724	0.4376	0	0.0165	0.1121	0.0295
8	0.4	1.5479	0.3733	0	0.0172	0.1258	0.0361
9	0.45	1.6056	0.3129	0	0.0176	0.1386	0.0435
10	0.5	1.6465	0.2579	0	0.0177	0.1503	0.0515
11	0.55	1.6731	0.2094	0.0001	0.0175	0.1604	0.0601
12	0.6	1.6884	0.1677	0.0001	0.0171	0.169	0.0692
13	0.65	1.6958	0.1328	0.0003	0.0165	0.1758	0.0787
14	0.7	1.6985	0.104	0.0006	0.0158	0.1811	0.0885
15	0.75	1.6992	0.0807	0.0011	0.0149	0.1847	0.0987
16	0.8	1.7006	0.0621	0.0022	0.014	0.1869	0.1092
17	0.85	1.705	0.0474	0.0043	0.0131	0.1876	0.1198
18	0.9	1.7148	0.0359	0.0083	0.0121	0.1869	0.1304
19	0.95	1.7321	0.0269	0.0159	0.011	0.1845	0.1407
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: South Korea → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.4587	0.3051	0.2852	0.0108	0.3018	0.0069
1	0.05	1.4962	0.2857	0.2735	0.0116	0.3157	0.0086
2	0.1	1.5337	0.2662	0.2609	0.0123	0.3285	0.0107
3	0.15	1.571	0.2467	0.2475	0.013	0.3398	0.0133

4	0.2	1.6078	0.2271	0.2334	0.0137	0.3495	0.0164
5	0.25	1.6437	0.2078	0.2186	0.0143	0.357	0.02
6	0.3	1.6783	0.1887	0.2033	0.0148	0.3621	0.0243
7	0.35	1.7111	0.1702	0.1877	0.0153	0.3646	0.0293
8	0.4	1.7416	0.1523	0.172	0.0156	0.3643	0.035
9	0.45	1.7693	0.1351	0.1563	0.0158	0.361	0.0416
10	0.5	1.7935	0.1189	0.1408	0.0158	0.3547	0.0489
11	0.55	1.8138	0.1037	0.1257	0.0158	0.3455	0.057
12	0.6	1.8296	0.0896	0.1112	0.0156	0.3334	0.0658
13	0.65	1.8404	0.0767	0.0975	0.0152	0.3188	0.0754
14	0.7	1.8458	0.0651	0.0847	0.0147	0.3021	0.0855
15	0.75	1.8456	0.0547	0.0729	0.0141	0.2835	0.0961
16	0.8	1.8395	0.0455	0.0621	0.0134	0.2636	0.1069
17	0.85	1.8276	0.0376	0.0525	0.0126	0.2428	0.1179
18	0.9	1.81	0.0307	0.0439	0.0118	0.2217	0.1289
19	0.95	1.7869	0.0249	0.0365	0.0109	0.2006	0.1396
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: Sweden → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.3772	0	0.0006	0.0115	0.276	0.3998
1	0.05	1.3969	0	0.0008	0.0117	0.2753	0.388
2	0.1	1.4167	0	0.001	0.0118	0.2743	0.376
3	0.15	1.4366	0	0.0012	0.0119	0.273	0.364
4	0.2	1.4566	0	0.0015	0.012	0.2713	0.3518
5	0.25	1.4766	0	0.0018	0.0121	0.2692	0.3396
6	0.3	1.4967	0	0.0022	0.0121	0.2667	0.3273
7	0.35	1.5166	0	0.0027	0.0122	0.2638	0.3149
8	0.4	1.5364	0	0.0034	0.0122	0.2604	0.3025
9	0.45	1.5558	0	0.0041	0.0122	0.2567	0.2899
10	0.5	1.5749	0	0.005	0.0122	0.2524	0.2773
11	0.55	1.5934	0	0.006	0.0121	0.2477	0.2647
12	0.6	1.6113	0.0001	0.0073	0.012	0.2425	0.2521
13	0.65	1.6285	0.0001	0.0089	0.0119	0.2367	0.2394
14	0.7	1.6449	0.0003	0.0107	0.0118	0.2305	0.2268
15	0.75	1.6607	0.0006	0.0128	0.0116	0.2237	0.2141
16	0.8	1.6761	0.0012	0.0154	0.0114	0.2164	0.2015
17	0.85	1.6921	0.0025	0.0183	0.0111	0.2086	0.1888
18	0.9	1.7098	0.005	0.0218	0.0108	0.2	0.1761
19	0.95	1.7313	0.01	0.0257	0.0104	0.1906	0.1633
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: United Kingdom → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.6572	0.0011	0.3105	0.0436	0.1243	0.019

1	0.05	1.6802	0.0013	0.2855	0.0419	0.1308	0.0217
2	0.1	1.7003	0.0016	0.2615	0.0401	0.1372	0.0248
3	0.15	1.7178	0.0019	0.2386	0.0382	0.1433	0.0282
4	0.2	1.7326	0.0022	0.2169	0.0362	0.1492	0.032
5	0.25	1.7451	0.0026	0.1965	0.0343	0.1547	0.0361
6	0.3	1.7553	0.0031	0.1773	0.0323	0.1599	0.0406
7	0.35	1.7636	0.0036	0.1595	0.0303	0.1646	0.0455
8	0.4	1.7701	0.0042	0.1429	0.0284	0.1689	0.0508
9	0.45	1.775	0.0048	0.1276	0.0265	0.1728	0.0566
10	0.5	1.7784	0.0056	0.1136	0.0246	0.1761	0.0628
11	0.55	1.7806	0.0064	0.1008	0.0228	0.1789	0.0694
12	0.6	1.7817	0.0074	0.0892	0.021	0.1812	0.0765
13	0.65	1.7818	0.0085	0.0786	0.0194	0.1829	0.0841
14	0.7	1.7809	0.0096	0.0691	0.0178	0.1841	0.0922
15	0.75	1.7792	0.011	0.0606	0.0163	0.1847	0.1007
16	0.8	1.7767	0.0124	0.0529	0.0148	0.1848	0.1097
17	0.85	1.7733	0.0141	0.0461	0.0135	0.1844	0.1191
18	0.9	1.7692	0.0159	0.0401	0.0123	0.1834	0.129
19	0.95	1.7643	0.0178	0.0347	0.0111	0.182	0.1393
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Geodesic Path: United States → Kardashev I

Step	t	Entropy	Coal	Gas	Oil	Nuclear	Hydro
0	0	1.6155	0.1491	0.4276	0.007	0.1788	0.0542
1	0.05	1.6677	0.1425	0.3957	0.0075	0.189	0.0603
2	0.1	1.7148	0.1354	0.3638	0.008	0.1985	0.0666
3	0.15	1.7561	0.1277	0.3322	0.0085	0.2071	0.0731
4	0.2	1.7913	0.1196	0.3013	0.009	0.2146	0.0797
5	0.25	1.82	0.1113	0.2715	0.0094	0.221	0.0863
6	0.3	1.8423	0.1029	0.243	0.0098	0.2259	0.0928
7	0.35	1.8583	0.0946	0.2161	0.0101	0.2296	0.0992
8	0.4	1.8686	0.0863	0.191	0.0104	0.2318	0.1053
9	0.45	1.8735	0.0783	0.1678	0.0106	0.2327	0.1112
10	0.5	1.8738	0.0707	0.1466	0.0108	0.2322	0.1167
11	0.55	1.8701	0.0635	0.1274	0.0109	0.2306	0.1219
12	0.6	1.8632	0.0567	0.1102	0.011	0.2278	0.1267
13	0.65	1.8538	0.0504	0.0949	0.011	0.224	0.131
14	0.7	1.8423	0.0446	0.0813	0.011	0.2194	0.135
15	0.75	1.8295	0.0394	0.0694	0.0109	0.214	0.1385
16	0.8	1.8158	0.0346	0.0591	0.0108	0.208	0.1416
17	0.85	1.8016	0.0303	0.0501	0.0106	0.2015	0.1443
18	0.9	1.7872	0.0264	0.0423	0.0104	0.1946	0.1466
19	0.95	1.7728	0.023	0.0357	0.0102	0.1874	0.1485
20	1	1.7587	0.02	0.03	0.01	0.18	0.15

Solar	Wind	Biofuel
0.1783	0.1162	0.0106
0.1948	0.1298	0.0128
0.2113	0.1438	0.0154
0.2274	0.1581	0.0183
0.2429	0.1725	0.0216
0.2575	0.1868	0.0253
0.2709	0.2009	0.0294
0.2831	0.2144	0.0339
0.2937	0.2272	0.0389
0.3027	0.2392	0.0443
0.31	0.2503	0.0502
0.3156	0.2604	0.0565
0.3196	0.2694	0.0632
0.322	0.2773	0.0704
0.3227	0.2839	0.0779
0.3217	0.2891	0.0859
0.3186	0.2925	0.094
0.3125	0.2931	0.1019
0.3016	0.289	0.1087
0.2825	0.2765	0.1125
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0956	0.1455	0.0781
0.103	0.1534	0.0815
0.1107	0.1614	0.0849
0.1186	0.1694	0.0883
0.1268	0.1773	0.0915
0.1351	0.1851	0.0945
0.1437	0.1927	0.0974
0.1523	0.2	0.1002
0.161	0.2071	0.1027
0.1697	0.2138	0.1049
0.1784	0.22	0.1069
0.187	0.2258	0.1087
0.1954	0.2311	0.1101
0.2035	0.2357	0.1112
0.2114	0.2398	0.112
0.219	0.2432	0.1125
0.2262	0.246	0.1126
0.2329	0.248	0.1124
0.2391	0.2494	0.1119

0.2448	0.25	0.1111
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0832	0.0988	0.0207
0.0949	0.1118	0.0243
0.1072	0.1252	0.0282
0.1198	0.1387	0.0324
0.1324	0.1519	0.0369
0.1448	0.1648	0.0415
0.1568	0.1769	0.0462
0.1682	0.1882	0.0511
0.1789	0.1984	0.0559
0.1889	0.2077	0.0607
0.198	0.2158	0.0655
0.2062	0.2229	0.0702
0.2137	0.2289	0.0748
0.2203	0.234	0.0794
0.2263	0.2383	0.0839
0.2316	0.2418	0.0883
0.2362	0.2445	0.0927
0.2404	0.2467	0.0971
0.244	0.2482	0.1014
0.2472	0.2493	0.1057
0.25	0.25	0.11

Solar	Wind	Biofuel
0.056	0.0776	0.0194
0.0624	0.085	0.0219
0.0693	0.0929	0.0246
0.0768	0.1012	0.0276
0.0847	0.1099	0.0308
0.0932	0.1189	0.0343
0.1021	0.1283	0.038
0.1115	0.1378	0.042
0.1214	0.1475	0.0463
0.1315	0.1573	0.0508
0.1421	0.1672	0.0555
0.1528	0.1769	0.0604
0.1638	0.1865	0.0655
0.1748	0.1959	0.0708
0.1859	0.205	0.0762
0.197	0.2137	0.0817

0.208	0.222	0.0873
0.2189	0.2298	0.0929
0.2295	0.2371	0.0986
0.2399	0.2438	0.1043
0.25	0.25	0.11

Solar	Wind	Biofuel
0.1802	0.2674	0.1016
0.1895	0.2758	0.1055
0.1988	0.2836	0.1093
0.2078	0.2907	0.1129
0.2167	0.2971	0.1163
0.2252	0.3028	0.1194
0.2335	0.3078	0.1222
0.2414	0.312	0.1248
0.2489	0.3154	0.1271
0.256	0.3181	0.1291
0.2627	0.32	0.1309
0.269	0.3212	0.1323
0.2747	0.3217	0.1335
0.28	0.3215	0.1344
0.2846	0.3204	0.1349
0.2885	0.3184	0.1351
0.2912	0.3151	0.1346
0.2917	0.3095	0.1332
0.288	0.2996	0.1299
0.2762	0.2817	0.123
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0674	0.0402	0.011
0.0807	0.0494	0.0138
0.0954	0.0599	0.0172
0.1114	0.0718	0.021
0.1281	0.0847	0.0254
0.1451	0.0984	0.0303
0.1618	0.1127	0.0355
0.1778	0.127	0.041
0.1925	0.1412	0.0466
0.2057	0.1548	0.0524
0.2172	0.1677	0.0581
0.2268	0.1797	0.0638
0.2346	0.1908	0.0693

0.2407	0.2009	0.0748
0.2453	0.2101	0.0801
0.2485	0.2184	0.0853
0.2506	0.2259	0.0904
0.2516	0.2328	0.0953
0.2517	0.239	0.1003
0.2512	0.2448	0.1052
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0038	0.0013	0.0601
0.0053	0.0019	0.07
0.0073	0.0028	0.081
0.0101	0.0041	0.093
0.0137	0.0059	0.1059
0.0185	0.0084	0.1195
0.0247	0.0118	0.1334
0.0327	0.0165	0.1474
0.0427	0.0226	0.1609
0.055	0.0308	0.1733
0.0698	0.0412	0.1841
0.0874	0.0543	0.1926
0.1076	0.0705	0.1983
0.1302	0.09	0.2006
0.1547	0.1127	0.1993
0.1803	0.1385	0.1942
0.2058	0.1666	0.1853
0.229	0.1955	0.1724
0.2472	0.2225	0.1556
0.256	0.2429	0.1347
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0951	0.0114	0.0447
0.1071	0.0142	0.0501
0.1196	0.0177	0.0558
0.1326	0.0218	0.0617
0.1459	0.0266	0.0676
0.1592	0.0323	0.0736
0.1724	0.0389	0.0794
0.1851	0.0465	0.085
0.1972	0.0551	0.0902
0.2084	0.0647	0.095

0.2186	0.0755	0.0994
0.2276	0.0875	0.1031
0.2354	0.1006	0.1063
0.2419	0.1149	0.1089
0.247	0.1305	0.1108
0.2507	0.1473	0.1121
0.2531	0.1654	0.1128
0.2541	0.1847	0.1129
0.2539	0.2053	0.1125
0.2525	0.2271	0.1115
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0025	0	0.0015
0.0036	0	0.0021
0.0051	0	0.0029
0.0072	0	0.0041
0.0101	0	0.0057
0.0141	0	0.0078
0.0196	0	0.0107
0.027	0	0.0146
0.0369	0	0.0196
0.0499	0.0001	0.0261
0.0667	0.0001	0.0343
0.088	0.0003	0.0445
0.1142	0.0008	0.0569
0.1456	0.0019	0.0714
0.1817	0.0044	0.0877
0.221	0.0099	0.1051
0.2599	0.0217	0.1217
0.292	0.0453	0.1346
0.307	0.0886	0.1393
0.2941	0.158	0.1314
0.25	0.25	0.11

Solar	Wind	Biofuel
0.1137	0.137	0.0493
0.1281	0.1529	0.0556
0.1432	0.1693	0.0622
0.1587	0.186	0.069
0.1745	0.2025	0.0759
0.1901	0.2186	0.0828
0.2054	0.234	0.0895

0.2201	0.2484	0.096
0.234	0.2616	0.1021
0.2469	0.2735	0.1078
0.2586	0.2838	0.113
0.2691	0.2926	0.1176
0.2782	0.2997	0.1217
0.2859	0.3051	0.1252
0.292	0.3088	0.1279
0.2965	0.3106	0.13
0.2988	0.3101	0.1311
0.2981	0.3065	0.1309
0.2924	0.2979	0.1285
0.2781	0.2807	0.1223
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0797	0.0446	0.0019
0.0968	0.0557	0.0027
0.1162	0.0689	0.0037
0.1377	0.084	0.0051
0.1608	0.101	0.0068
0.1846	0.1194	0.0091
0.2084	0.1387	0.0119
0.2309	0.1582	0.0152
0.2512	0.1773	0.0191
0.2686	0.1951	0.0237
0.2825	0.2113	0.0288
0.2926	0.2253	0.0346
0.299	0.237	0.0409
0.3019	0.2463	0.0478
0.3016	0.2533	0.0552
0.2985	0.2581	0.0633
0.293	0.2608	0.0719
0.2853	0.2615	0.081
0.2757	0.2602	0.0906
0.2641	0.2565	0.1004
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0523	0.0054	0.0325
0.0607	0.007	0.0371
0.0701	0.009	0.0421
0.0805	0.0116	0.0475

0.0918	0.0149	0.0533
0.1041	0.0189	0.0594
0.1172	0.0238	0.0657
0.131	0.0298	0.0721
0.1452	0.0371	0.0786
0.1596	0.0457	0.0849
0.1741	0.0558	0.091
0.1881	0.0675	0.0967
0.2015	0.0811	0.1018
0.2138	0.0964	0.1062
0.2247	0.1135	0.1097
0.234	0.1325	0.1123
0.2414	0.1531	0.1138
0.2468	0.1754	0.1144
0.25	0.1991	0.1139
0.2511	0.2241	0.1124
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0267	0.227	0.0584
0.0304	0.2324	0.0614
0.0347	0.2377	0.0645
0.0394	0.2429	0.0677
0.0447	0.2478	0.0709
0.0507	0.2524	0.0742
0.0574	0.2567	0.0775
0.0649	0.2606	0.0809
0.0732	0.2642	0.0842
0.0824	0.2672	0.0875
0.0926	0.2698	0.0907
0.1038	0.2718	0.0939
0.116	0.2731	0.0969
0.1294	0.2737	0.0998
0.144	0.2736	0.1025
0.1597	0.2726	0.1049
0.1765	0.2707	0.107
0.1943	0.2678	0.1087
0.2128	0.2636	0.1099
0.2317	0.2579	0.1105
0.25	0.25	0.11

Solar	Wind	Biofuel
0.066	0.2948	0.1408

0.0729	0.3022	0.1437
0.0802	0.3085	0.1461
0.0879	0.3138	0.148
0.096	0.318	0.1494
0.1045	0.3211	0.1502
0.1133	0.323	0.1505
0.1224	0.3238	0.1503
0.1318	0.3235	0.1495
0.1414	0.3221	0.1482
0.1512	0.3196	0.1465
0.1612	0.3161	0.1443
0.1713	0.3117	0.1417
0.1814	0.3064	0.1387
0.1916	0.3002	0.1354
0.2017	0.2933	0.1317
0.2118	0.2857	0.1278
0.2217	0.2775	0.1236
0.2314	0.2688	0.1192
0.2409	0.2596	0.1147
0.25	0.25	0.11

Solar	Wind	Biofuel
0.0695	0.1033	0.0106
0.0783	0.1141	0.0126
0.0876	0.1253	0.0149
0.0974	0.1365	0.0174
0.1076	0.1478	0.0203
0.118	0.159	0.0235
0.1286	0.1699	0.027
0.1393	0.1803	0.0308
0.1499	0.1903	0.035
0.1604	0.1995	0.0394
0.1706	0.2081	0.0442
0.1805	0.2158	0.0494
0.1901	0.2228	0.0548
0.1992	0.2289	0.0605
0.2079	0.2342	0.0666
0.2161	0.2387	0.073
0.2239	0.2424	0.0797
0.2312	0.2453	0.0868
0.2379	0.2476	0.0942
0.2442	0.2491	0.1019
0.25	0.25	0.11

Country	$\bar{\chi}$	W	CV(χ)	Regime	Gov Efficiency	K1 Distance
China	0.350969	0.369739	0.477	C-U	0.2376	4.2067
United States	0.332278	0.097675	0.6516	C-M	0.2012	4.4681
India	0.378048	0.426668	0.7538	C-M	0.2156	5.0513
Japan	0.449327	0.127716	0.5011	C-M	0.2993	5.1713
Germany	0.55129	0.056714	0.7572	B-M	0.3137	16.4335
United Kingdom	0.421085	0.033011	1.1238	C-B	0.1983	4.6309
France	0.433586	0.198433	0.6622	C-M	0.2609	3.1137
Brazil	0.467487	0.213593	0.7962	C-M	0.2603	2.9029
Russia	0.409761	0.022095	0.7126	C-M	0.2393	8.2153
Canada	0.412878	0.251945	0.5015	C-M	0.275	4.224
Australia	0.05451	0.091305	0.6682	A-M	0.0327	16.4198
South Korea	0.350969	0.188146	0.5548	C-M	0.2257	6.2303
Indonesia	0.3337	0.095755	0.9307	C-M	0.1728	16.8468
Mexico	0.36743	0.342179	0.6226	C-M	0.2264	5.01
South Africa	0.297025	0.067304	0.9172	C-M	0.1549	14.2403
Turkey	0.246363	0.135003	1.163	C-B	0.1139	16.2611
Italy	0.234979	0.136984	1.1352	C-B	0.1101	16.135
Spain	0.337017	0.104321	0.7567	C-M	0.1918	3.3795
Poland	0.252943	0.026823	0.8108	C-M	0.1397	16.5189
Sweden	0.351993	0.072833	1.1577	C-B	0.1631	13.5276
Norway	0.275003	0.117632	0.6505	C-M	0.1666	14.8895
Denmark	0.246457	0.032011	0.753	C-M	0.1406	16.2082
Finland	0.414148	0.304646	1.1434	C-B	0.1932	3.7773
Iran	0.378956	0.138855	0.8102	C-M	0.2093	9.2505
Nigeria	0.219846	0.128043	0.4382	C-U	0.1529	21.3218
Egypt	0.159076	0.072221	0.8747	C-M	0.0849	22.2652
Thailand	0.271213	0.13428	0.96	C-M	0.1384	16.1381
Vietnam	0.252593	0.132046	0.8535	C-M	0.1363	16.1346
Philippines	0.277688	0.085178	1.1295	C-B	0.1304	16.4138
Colombia	0.256606	0.088706	0.8532	C-M	0.1385	21.6281
Chile	0.268396	0.081541	1.4161	C-B	0.1111	16.3168
Argentina	0.482249	0.117873	0.8956	C-M	0.2544	4.5558
Pakistan	0.423321	0.060141	0.6976	C-M	0.2494	5.1181
Bangladesh	0.229598	0.099434	0.6982	C-M	0.1352	24.2212
Kenya	0.20494	0.056865	0.6122	C-M	0.1271	21.7979
Ethiopia	0.214851	0.060379	0.5999	C-M	0.1343	18.5748
Morocco	0.292601	0.056055	0.8829	C-M	0.1554	16.6347
Ukraine	0.392424	0.060886	0.9414	C-M	0.2021	5.0082
Israel	0.246562	0.010085	0.9562	C-M	0.126	20.6068

K1 ETA (yrs)	χ toward K1	Fossil %	Renewables %
21.1	0.6	61.83478165	33.69555283
35	0.4119	58.1306839	24.06342888
42.2	0.2675	77.74752045	19.55817223
5.9	0.5034	68.71083069	22.93509293
50.8	0.1842	41.15672302	58.84327698
25.3	0.2734	35.52496719	52.04932022
17.8	0.2791	5.16443491	25.84680939
4.8	0.6454	10.55087757	87.33305359
31.5	0.5073	64.0646286	18.09792328
26.9	0.437	22.30562592	64.27236938
1750.3	0.0303	64.9233017	35.07669067
32.7	0.4787	60.10905457	9.70929718
24.2	0.2323	81.91042328	18.08957291
19.9	0.3373	75.41809845	21.11605453
82.8	0.0673	83.81177521	13.05657005
273.5	0.073	56.74277878	43.25722504
238.9	0.1327	51.31708145	48.68291855
14.9	0.3009	25.2979908	55.97366333
323	0.0771	68.91805267	31.08194351
39.8	0.2027	1.218654752	71.18584442
53.6	0.1813	1.207608461	98.79238892
190.3	0.1267	8.797127724	91.20287323
6.4	0.6447	3.664084435	55.98180008
18	0.3006	92.82447815	5.318476677
1679.4	0.045	76.97483063	23.02516747
335.4	0.0764	88.80323792	11.19676495
230.5	0.0822	85.15938568	14.84061718
87.1	0.1189	57.71790314	42.28210068
143.9	0.1537	78.26950836	21.73049355
44.3	0.1266	35.64163971	64.35836029
274	0.1075	30.10306931	69.89692688
10.8	0.4355	58.76336288	34.38463211
14.1	0.397	53.64085388	32.90499496
128.1	0.0855	97.97119141	2.028807163
469.2	0.05	8.093384743	91.90661621
18.6	0.2879	0	100
43.8	0.1695	73.49589539	26.5041008
11.8	0.4307	27.82710075	16.41108322
25.8	0.1395	89.49065399	10.50934029

Carbon gCO2/kWh	Stagnation $M/yr	GDP/cap $	Stage	Coal%
555.4000244	280112.7	18999	Early transition	0.5777
383.7799988	84259.3	56432	Mid-transition	0.1491
707.4500122	71813.2	7220	Early transition	0.7477
483.4299927	24567.7	38581	Early transition	0.3219
331.6000061	8247.6	46501	Mid-transition	0.2074
217.0899963	3178	37601	Active transition	0.0011
41.79000092	1187.5	40083	Post-fossil (K1 approaching)	0.0031
106.0599976	3954.6	15035	Post-fossil (K1 approaching)	0.0217
445.980011	26966.4	25766	Early transition	0.1779
185.3500061	5901.5	44318	Active transition	0.0395
553.8300171	7799	50322	Early transition	0.4521
415.5100098	12992.6	41331	Early transition	0.3051
680.25	12637	12347	Early transition	0.6439
483.1400146	8580.1	15539	Early transition	0.0349
713.1699829	8881.5	11304	Early transition	0.8303
474.7399902	8399.6	27211	Early transition	0.3537
285.2600098	3768.6	36114	Early transition	0.0144
153.2599945	2211.5	33935	Mid-transition	0.0032
592.2000122	5098.5	32276	Early transition	0.5078
35.33000183	301.5	46376	Post-fossil (K1 approaching)	0
29.10000038	232.5	85777	Post-fossil (K1 approaching)	0.0002
114.3000031	191	49884	Post-fossil (K1 approaching)	0.0266
56.81000137	231	40217	Post-fossil (K1 approaching)	0.002
648.6799927	12837.1	17170	Degrading (moving away fro	0.0018
507.8500061	1019	4876	Early transition	0
574.5	6819	12532	Pre-transition (fossil locked	0
554.7299805	5534	15685	Early transition	0.1667
484.3200073	7351.5	8256	Early transition	0.5032
612.3499756	3853.5	8760	Early transition	0.6807
285.7999878	1250.5	14044	Early transition	0.1515
265.1799927	1183.5	22684	Early transition	0.1585
344.8299866	2629.5	18709	Mid-transition	0.0086
398.2399902	3626	5249	Mid-transition	0.1395
693.8400269	3950	4944	Early transition	0.2016
84.83000183	54.5	3373	Post-fossil (K1 approaching)	0
23.54999924	21.5	2098	Post-fossil (K1 approaching)	0
576.5700073	1265	8184	Early transition	0.5928
250.4700012	1396.5	7532	Mid-transition	0.2091
567.2600098	2110.5	36378	Early transition	0.1765

Gas%	Oil%	Nuclear%	Hydro%	Solar%	Wind%	Biofuel%
0.0318	0.0088	0.0447	0.1343	0.0832	0.0988	0.0207
0.4276	0.007	0.1788	0.0542	0.0695	0.1033	0.0106
0.0278	0.002	0.0269	0.0771	0.0674	0.0402	0.011
0.3407	0.0246	0.0835	0.0782	0.0951	0.0114	0.0447
0.1662	0.038	0	0.0393	0.1802	0.2674	0.1016
0.3105	0.0436	0.1243	0.019	0.066	0.2948	0.1408
0.0304	0.0182	0.6906	0.1046	0.056	0.0776	0.0194
0.0654	0.0184	0.0212	0.5541	0.0956	0.1455	0.0781
0.4513	0.0114	0.1784	0.1741	0.0024	0.0038	0.0007
0.1721	0.0115	0.1342	0.539	0.0148	0.0735	0.0154
0.1743	0.0228	0	0.0456	0.1783	0.1162	0.0106
0.2852	0.0108	0.3018	0.0069	0.0523	0.0054	0.0325
0.1939	0.0201	0	0.0769	0.0038	0.0013	0.0601
0.6241	0.0952	0.0347	0.0663	0.0724	0.0556	0.0169
0	0.0078	0.0313	0.0044	0.0797	0.0446	0.0019
0.2283	0.003	0	0.1669	0.1085	0.1149	0.0247
0.4837	0.0263	0	0.1596	0.1726	0.0829	0.0604
0.215	0.0348	0.1873	0.1134	0.2179	0.2061	0.0222
0.1443	0.038	0	0.0099	0.1137	0.137	0.0493
0.0006	0.0115	0.276	0.3998	0.0267	0.227	0.0584
0.0071	0.0048	0	0.8998	0.0023	0.0843	0.0016
0.0242	0.0371	0	0.0006	0.1332	0.5784	0.1999
0.0113	0.0234	0.4042	0.1501	0.0124	0.2734	0.1232
0.8592	0.0672	0.0186	0.0477	0.0023	0.0032	0.0001
0.7697	0	0	0.2263	0.0025	0	0.0015
0.8129	0.0751	0	0.0605	0.0256	0.0259	0
0.6835	0.0014	0	0.0321	0.0272	0.0169	0.0721
0.0719	0.002	0	0.2922	0.0852	0.042	0.0034
0.1536	0.018	0	0.094	0.0318	0.0106	0.0112
0.1683	0.0366	0	0.581	0.0367	0	0.0259
0.1164	0.0272	0	0.2969	0.2239	0.1189	0.0581
0.5373	0.0417	0.0685	0.1945	0.026	0.106	0.0172
0.2496	0.1473	0.1345	0.1887	0.0997	0.0299	0.0107
0.6154	0.1627	0	0.0078	0.0125	0	0
0	0.1427	0	0.4952	0.0754	0.2483	0.0384
0	0	0	0.9665	0.0022	0.0307	0.0005
0.101	0.0412	0	0.0153	0.0362	0.2126	0.0009
0.0644	0.0048	0.5576	0.0988	0.0456	0.0127	0.007
0.7139	0.0046	0	0	0.0937	0.0099	0.0015

HUF Kardashev Framework — Methodology

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Date: 2026-04-12

DIAGNOSTIC MEASURES

Metric	Formula
$\bar{\chi}$ (crossing intensity)	$\lvert v_\perp \rvert / \lvert v \rvert$ averaged
W (winding ratio)	crossings / Aitchison path
CV(χ) (uniformity)	$\sigma\_\chi / \bar{\chi}$ (Welford)
Governance Efficiency	$\bar{\chi} / (1 + \mathrm{CV}(\chi))$
Parallel Waste	$1 - \bar{\chi}$

KARDASHEV I TARGET

Definition	Civilisation harnessing $\sim 1.74\times10^{16}$ W (≈4× current g
Target Mix	Near-zero fossil: Coal 2%, Gas 3%, Oil 1%, Nuclear 1
K1 Geodesic Distance	Aitchison distance from current composition to K1
χ toward K1	cos(θ) between actual velocity and direction to K1
ETA K1 (years)	K1 distance / (avg speed × χ toward K1) — at curre

DOLLAR COST FRAMEWORK

Stagnation Cost	Elec generation (TWh) × Carbon intensity (gCO2/kW
Dollar Waste % GDP	Energy/GDP × Parallel Waste × 100

DEVELOPMENT STAGES

Post-fossil (K1 approaching)	Fossil < 20%
Active transition	Fossil < 40%, $\bar{\chi} > 0.4$
Mid-transition	Fossil < 60%, $\bar{\chi} > 0.3$
Early transition	Mixed indicators
Pre-transition (fossil locked)	Fossil > 80%, $\bar{\chi} < 0.2$
Degrading	Fossil > 60% and rising
Unstable	CV(χ) > 1.5

DATA SOURCES	
OWID Energy Data	ourworldindata.org — 211 countries, yearly, 1965-
Ember Global Monthly	ember-climate.org — major economies, monthly g
Ember Europe Monthly	ember-climate.org — 39 European countries, 11 fu

GEOMETRY	
Simplex	Compositions live on the D-simplex (shares sum to
Aitchison metric	Fisher-Rao / Čencov-canonical metric on compositi
ILR coordinates	Isometric log-ratio transform — compositions → E
Geodesic	Straight line in CLR space = shortest path on simple

Range	Interpretation
[0, 1]	Fraction of energy-mix motion crossing entropy contours (purposeful transition)
[0, ∞)	Frequency of entropy-level reversals
[0, ∞)	Temporal uniformity: 0=steady, >>1=burst-driven
[0, 1]	Rewards purposeful, consistent transition; penalises waste and inconsistency
[0, 1]	Fraction of motion along entropy contours — structural redistribution without entropy change

global primary energy)
18%, Hydro 15%, Solar 25%, Wind 25%, Biofuel 11%
target on the simplex
target in ILR space
nt pace and alignment

Wh) / 100(	Annual social cost of carbon at current electricity mix
	Fraction of economic output consumed by non-transitional energy-mix churning

-2025
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THE GOVERNANCE CASE FOR FUSION

From Stagnation Cost to Self-Bootstrapping K1 Technology

Peter Higgins — Rogue Wave Audio — 2026-04-12

1. THE PLANETARY STAGNATION COST

Metric	Value
Total annual stagnation cost	$707.3B
Global fusion research spend	$5–6B/yr
Stagnation-to-fusion ratio	141:1
Top 3 countries (China+US+India)	$436B
Top 15 countries	$585B
1% redirection of stagnation cost	$7.1B/yr
5% redirection	$35.4B/yr
10% redirection	$70.7B/yr

2. THE SELF-BOOTSTRAPPING MECHANISM

Each GW of fusion deployed REDUCES the stagnation cost that funds the next GW.

Parameter	Value
1 GW fusion capacity factor	90%
Annual generation per GW	7.884 TWh/yr
CO_2 displaced per GW (global avg)	3.55 $MtCO_2$/yr
Stagnation cost eliminated per GW	$177M/yr
At realistic SCC ($100/$tCO_2$)	$355M/yr per GW
Estimated first-gen build cost	$15–20B per GW
Social payback period ($50/t)	~85–113 years
Social payback period ($100/t)	~42–56 years

THE BOOTSTRAP LOOP:	
Step 1	Redirect 1% of stagnation cost ($7.1B/yr)
Step 2	Build 0.35–0.47 GW fusion/yr
Step 3	Each GW reduces stagnation cost by $177M/yr
Step 4	Freed stagnation cost funds MORE fusion
Step 5	Learning curve reduces build cost
Step 6	Loop accelerates — the bootstrap is self-sustaining

3. PER-COUNTRY FUSION DIVIDEND — What Each Nation Saves

Country	Stagnation Cost ($M/yr)
China	280113
United States	84259
India	71813
Russia	26966
Japan	24568
South Korea	12993
Iran	12837
Indonesia	12637
Taiwan	9164

South Africa	8882
Mexico	8580
Turkey	8400
Germany	8248
Australia	7799
Vietnam	7352
Egypt	6819
Malaysia	5990
Canada	5902
Thailand	5534
Iraq	5180
Poland	5098
Kazakhstan	4752
Uzbekistan	4274
Brazil	3955
Bangladesh	3950
Philippines	3854
Italy	3769
Pakistan	3626
United Kingdom	3178
Algeria	3051

4. WHY GOVERNANCE — NOT MARKETS — IS THE BOTTLENECK

Argument	Evidence / Reasoning
The Mathematical Path is Short	The Aitchison geodesic from any country's current mix to K1 is computable and finite. Brazil: 2.9 units. India: 5.1 units. The simplex geometry tells us the journey is not long.
The Governed Path is Long	Australia's K1 ETA: 1,750 years. Not because of distance (it's middling) but because its effective speed toward K1 is 0.009 — essentially zero alignment between motion and target.
The Gap is the Cost	The difference between the mathematical path and the governed path IS the governance cost. It is measurable in dollars per year. It is $707B globally.
Markets Cannot Self-Correct	The parallel waste fraction (55–100% across nations) represents energy-mix motion that markets drive but that produces zero entropy transition. Markets optimise for price, not for compositional direction on the simplex.
Only Governance Can Set Direction	χ toward K1 is a POLICY variable. It measures whether national energy decisions point toward the target. It requires coordinated direction-setting that markets do not provide.
Fusion Changes the Simplex	Current K1 target assumes redistribution of existing clean sources. Fusion adds a new vertex to the simplex — a fuel type with effectively unlimited capacity. This is not redistribution; it is dimensional expansion.

Concentration Makes Coordination Tractable	82.6% of global stagnation cost sits in 15 countries. This is not a 200-country coordination problem. It is a 15-seat table.
The Money Already Exists	This is not a funding problem. $707B/yr is already being spent — on stagnation. The governance question is: redirect existing waste, not find new money.
Developing Nations Leapfrog	Fusion allows developing nations to skip the fossil lock-in phase entirely. Countries currently at 'Pre-transition (fossil locked)' stage can jump directly to post-fossil if fusion becomes available as infrastructure.
Energy Security as Driver	Fusion eliminates fuel-supply dependency. For nations currently locked into gas/coal imports, fusion is not just a climate technology — it is a sovereignty technology. Every nation with a tokamak is energy-independent.
The Diagnostic is Already Built	The HUF framework provides the measurement instrument: $\bar{\chi}$ for transition effort, CV(χ) for consistency, χ toward K1 for alignment, stagnation cost for dollar accountability. Governance has a dashboard.
Stagnation Cost is Self-Evident Funding	Every dollar of stagnation cost is a dollar the planet is already paying for carbon it doesn't need to emit. Redirecting it to fusion doesn't increase expenditure — it transmutes punishment into investment.

5. FUSION TIMELINE SCENARIOS — FROM GOVERNANCE TO K1

Scenario	Annual Redirect
Status Quo	$5B (current)
1% Redirect	$7.1B/yr
5% Redirect	$35B/yr
10% Redirect	$71B/yr
10% + Learning Curve	$71B→$140B/yr
Full Bootstrap (theoretical)	$707B/yr→$0

6. THE FEEDBACK EQUATION

Variable	Formula
Stagnation Cost(t)	S(t) = Σ_countries [Elec_i(t) × CI_i(t) × SCC]
Fusion Deployed(t)	F(t) = F(t-1) + Investment(t) / BuildCost(t)
Carbon Displaced(t)	ΔC(t) = F(t) × CF × 8760 × CI_avg / 1000
Stagnation Reduction(t)	ΔS(t) = ΔC(t) × SCC
Available for Reinvestment(t)	R(t) = α × S(t) + ΔS(t)
Build Cost Learning(t)	$BC(t) = BC_0 \times (F(t)/F_0)$^(-ln2/ln(1/LR))
K1 Gap(t)	G(t) = E_K1 - E_current(t) - E_fusion(t)
Bootstrap Condition	R(t) > BC(t) for all t > t*
K1 Arrival	t_K1 : G(t_K1) = 0

7. THE COMPLETE VISION: GOVERNANCE → MEASUREMENT → FUSION → K1

The HUF decomposition framework provides three things that did not previously exist in combination:

1. A MEASUREMENT INSTRUMENT — $\bar{\chi}$, W, CV(χ) — that quantifies the quality of energy transi
2. A DOLLAR FIGURE — the stagnation cost — that converts governance failure into an annual
3. A TARGET — the Kardashev I energy mix — expressed as a point on the compositional simp

The fusion case follows directly:

- The stagnation cost is real money already being spent. It is $707B/yr. It funds nothing productive.
- Fusion is the technology that collapses the K1 timeline from centuries to decades.
- The self-bootstrap mechanism means initial governance redirection creates its own funding stream.
- The HUF framework provides the diagnostic dashboard to monitor progress: are we moving toward K

This is not a proposal for new spending. It is a proposal for redirecting existing waste toward the one te that makes waste itself unnecessary. The mathematics are on the simplex. The governance is at the ta The money is already in the room.

Interpretation

Social cost of carbon from current electricity mix at $50/t$CO_2$

ITER, national programs, private ventures combined

Planet spends 141× more on NOT transitioning than on the technology that WOULD

61.7% of global stagnation cost — three governance decisions

82.6% of global cost — a tractable coordination problem

Would MORE THAN DOUBLE current global fusion investment

Would match Apollo program scale ($35B inflation-adjusted per year)

Manhattan-to-Apollo scale sustained — comparable to global military R&D

Source / Derivation

ITER/DEMO design target

1 GW × 8,760 hrs × 0.90

7.884 TWh × 450 gCO_2/kWh ÷ 1000

3.55 Mt × $50/t$CO_2$

3.55 Mt × $100/t$CO_2$

ITER extrapolation, learning curve TBD

Build cost ÷ annual stagnation savings

Faster at realistic carbon pricing

Governance decision, not new money

At $15–20B/GW

Carbon displaced from grid

Positive feedback loop

Every doubling ≈ 20% cost reduction (historical energy tech)

The mathematical path becomes the governed path

% of GDP	1% Redirect ($M)
1.039	2801.1
0.432	842.6
0.685	718.1
0.723	269.7
0.515	245.7
0.608	129.9
0.817	128.4
0.361	126.4
0.734	91.6

1.227	88.8
0.422	85.8
0.352	84
0.211	82.5
0.58	78
0.882	73.5
0.467	68.2
0.683	59.9
0.335	59
0.492	55.3
0.982	51.8
0.414	51
0.944	47.5
1.086	42.7
0.124	39.5
0.46	39.5
0.38	38.5
0.176	37.7
0.275	36.3
0.122	31.8
0.512	30.5

Source
K1 Path Analysis sheet — geodesic distances
χ toward K1 = 0.030 for Australia
Stagnation Cost = Elec × CI × $50/t$CO_2$
Parallel waste is structural, not inefficiency
χ toward K1 ranges from 0.03 (Australia) to 0.65 (Brazil)
Fusion = new simplex dimension

Top 15 countries = $585B of $707B

Redirect, not raise

See Developing Nations sheet — 30+ countries with GDP/cap < $20k

Geopolitical: removes resource leverage

This spreadsheet IS the diagnostic tool

Reframe: not spending more, spending differently

Fusion GW/yr Built	Years to 1 TW Fusion
~0 (R&D only)	Never
0.4 GW/yr	~2,500 yrs
2 GW/yr	~500 yrs
4 GW/yr	~250 yrs
4→20 GW/yr	~80 yrs
All stagnation→fusion	~40 yrs

Meaning
Total global stagnation cost at time t
Cumulative fusion capacity
Annual CO_2 avoided by fusion fleet
Stagnation cost eliminated by fusion
α = governance redirect fraction + freed stagnation
LR = learning rate, typically 0.80 for energy tech
Remaining energy gap to Kardashev I
Self-sustaining when annual savings exceed annual build cost
The year the planet reaches Kardashev I

tion, not just its existence.
bill, per country, per year.
lex with a computable geodesic path from every nation's current position.

:1? How fast? How efficiently?

echnology
able.

5% Redirect ($M)	GW Fundable/yr @1%	Fossil Share %	K1 ETA (years)
14005.6	0.1601	62	21
4213	0.0481	58	35
3590.7	0.041	78	42
1348.3	0.0154	64	32
1228.4	0.014	69	6
649.6	0.0074	60	33
641.9	0.0073	93	18
631.9	0.0072	82	24
458.2	0.0052	85	30

444.1	0.0051	84	83
429	0.0049	75	20
420	0.0048	57	274
412.4	0.0047	41	51
390	0.0045	65	1750
367.6	0.0042	58	87
341	0.0039	89	335
299.5	0.0034	80	617
295.1	0.0034	22	27
276.7	0.0032	85	230
259	0.003	99	274
254.9	0.0029	69	323
237.6	0.0027	85	19
213.7	0.0024	91	91
197.7	0.0023	11	5
197.5	0.0023	98	128
192.7	0.0022	78	144
188.4	0.0022	51	239
181.3	0.0021	54	14
158.9	0.0018	36	25
152.6	0.0017	99	51

CO_2 Displaced at 1 TW	K1 Acceleration Factor	Notes
0	1.0×	Current trajectory — K1 unreachable
3.5 Mt/yr per GW	1.4×	Doubles fusion funding — meaningful R&D acceleration
17.5 Mt/yr per GW	7×	Apollo-scale — serious engineering program
35 Mt/yr per GW	14×	Manhattan-scale sustained — transforms the timeline
Exponential	50×+	Learning curve halves cost every decade — the bootstrap ac
All electricity	∞	Theoretical limit: all stagnation cost becomes fusion investm

ccelerates
ent until K1

FUSION BOOTSTRAP PROJECTION MODEL

Assumptions

Parameter	Value	Source / Note
Social Co	50	Source: Interagency Working Group, conservative estim
Global Av	450	Source: OWID/Ember 2024 global average
Fusion Ca	0.9	Source: ITER/DEMO design specification
Initial Buil	20	Source: ITER extrapolation
Learning I	0.8	20% cost reduction per capacity doubling
Governan	0.05	5% of stagnation cost redirected to fusion
Global Sta	707.3	HUF framework computation from OWID data
K1 Target	17.4	Kardashev I = 1.74×10^{16} W
Current G	3.1	~27,000 TWh/yr ÷ 8760

50-YEAR BOOTSTRAP PROJECTION

Year	Stagnation Cost ($B)	Redirect ($B)	Build Cost ($/GW, $B)
2026	707.3	35.365	20
2027	706.9863301	35.34931651	20
2028	706.3562083	35.31781042	15.977224
2029	705.3266183	35.26633091	13.64122723
2030	703.8243681	35.1912184	12.07903221
2031	701.7821426	35.08910713	10.94206658
2032	699.136298	34.9568149	10.06688512
2033	695.8255205	34.79127602	9.365918421
2034	691.7899568	34.58949784	8.787652145
2035	686.9706237	34.34853118	8.299611378
2036	681.3089942	34.06544971	7.880203611
2037	674.7467024	33.73733512	7.514451938
2038	667.2253314	33.36126657	7.191604825
2039	658.6862623	32.93431312	6.903720928
2040	649.0705708	32.45352854	6.644792104
2041	638.3189604	31.91594802	6.410178728
2042	626.3717266	31.31858633	6.196234049
2043	613.1687479	30.65843739	6.000047184
2044	598.6494994	29.93247497	5.819262937
2045	582.7530885	29.13765443	5.651952716
2046	565.41831	28.2709155	5.496520304
2047	546.5837199	27.329186	5.351631875
2048	526.1877284	26.30938642	5.216163233
2049	504.1687108	25.20843554	5.089159485
2050	480.4651368	24.02325684	4.969803822
2051	455.0157191	22.75078595	4.857393094
2052	427.7595795	21.38797898	4.751318481
2053	398.6364365	19.93182182	4.651050075
2054	367.5868106	18.37934053	4.556124452
2055	334.5522523	16.72761262	4.466134593
2056	299.47559	14.9737795	4.380721639

2057	262.3012002	13.11506001	4.299568108
2058	222.9753004	11.14876502	4.222392271
2059	181.4462651	9.072313257	4.148943477
2060	137.6649662	6.883248309	4.078998238
2061	91.58513662	4.579256831	4.012356934
2062	43.16376054	2.158188027	3.948841038
2063	0	0	3.888290765
2064	0	0	3.830563077
2065	0	0	3.775143882
2066	0	0	3.719748596
2067	0	0	3.664377896
2068	0	0	3.609032487
2069	0	0	3.553713105
2070	0	0	3.498420519
2071	0	0	3.443155529
2072	0	0	3.387918975
2073	0	0	3.332711732
2074	0	0	3.277534718
2075	0	0	3.222388894
2076	0	0	3.167275266

ıate

Fusion Built (GW/yr)	Cumulative Fusion (GW)	Fusion Gen (TWh/yr)
1.76825	1.76825	13.940883
1.783933493	3.552183493	28.00541466
2.251920505	5.804103999	45.75955592
2.664525691	8.46862969	66.76667647
3.044000588	11.51263028	90.76557711
3.40277987	14.91541015	117.5930936
3.748423796	18.66383394	147.1456668
4.085834478	22.74966842	179.3583858
4.418340538	27.16800896	214.1925826
4.748274245	31.9162832	251.6279768
5.077300364	36.99358357	291.6574129
5.40661406	42.40019763	334.2831581
5.737065256	48.13726288	379.5141806
6.069239483	54.20650237	427.3640647
6.403511789	60.61001415	477.8493516
6.740083356	67.35009751	530.9881688
7.079006615	74.42910413	586.7990569
7.420202499	81.84930663	645.2999334
7.763472176	89.6127788	706.5071481
8.108504812	97.72128361	770.4346
8.454882426	106.176166	837.0928931
8.80208256	114.9782486	906.488512
9.149479297	124.1277279	978.6230067
9.496343004	133.6240709	1053.492175
9.84183908	143.46591	1131.085234
10.18502593	153.6509359	1211.383979
10.52485235	164.1757883	1294.361915
10.86015443	175.0359427	1379.983372
11.18965213	186.2255948	1468.20259
11.51194567	197.7375405	1558.962769
11.82551168	209.5630522	1652.195103

12.12869945	221.6917516	1747.81777
12.41972713	234.1114787	1845.734898
12.69667821	246.808157	1945.835509
12.95749817	259.7656551	2047.992425
13.19999161	272.9656467	2152.061159
13.42181982	286.3874666	2257.878786
13.62049906	300.0079656	2365.262801
13.89310448	313.9010701	2474.796037
14.74987777	328.6509479	2591.084073
15.67293868	344.3238866	2714.649522
16.66848125	360.9923678	2846.063828
17.74338035	378.7357482	2985.952638
18.90527805	397.6410262	3135.001851
20.16268235	417.8037085	3293.964438
21.52508048	439.328789	3463.668173
23.00306898	462.331858	3645.024369
24.60850349	486.9403615	3839.03781
26.35467147	513.295033	4046.81804
28.25649197	541.5515249	4269.592223
30.33074708	571.882272	4508.719833

CO_2 Displaced (Mt/yr)	Stagnation Saved ($B/yr)	Total Available ($B)
6.27339735	0.313669868	35.67866987
12.6024366	0.63012183	35.97943834
20.59180017	1.029590008	36.34740042
30.04500441	1.502250221	36.76858114
40.8445097	2.042225485	37.23344389
52.91689212	2.645844606	37.73495174
66.21555006	3.310777503	38.2675924
80.71127362	4.035563681	38.82683971
96.38666218	4.819333109	39.40883095
113.2325896	5.661629478	40.01016066
131.2458358	6.562291789	40.6277415
150.4274211	7.521371057	41.25870618
170.7813813	8.539069063	41.90033563
192.3138291	9.615691455	42.55000457
215.0322082	10.75161041	43.20513895
238.9446759	11.9472338	43.86318182
264.0595756	13.20297878	44.52156511
290.38497	14.5192485	45.1776859
317.9282166	15.89641083	45.8288858
346.69557	17.3347785	46.47243293
376.6918019	18.83459009	47.10550559
407.9198304	20.39599152	47.72517752
440.380353	22.01901765	48.32840407
474.0714787	23.70357394	48.91200948
508.9883554	25.44941777	49.47267461
545.1227904	27.25613952	50.00692547
582.4628616	29.12314308	50.51112206
620.9925175	31.04962587	50.9814477
660.6911653	33.03455827	51.41389879
701.5332462	35.07666231	51.80427492
743.4877965	37.17438983	52.14816933

786.5179964	39.32589982	52.44095983
830.5807043	41.52903522	52.67780023
875.6259792	43.78129896	52.85361222
921.5965913	46.07982956	52.96307787
968.4275215	48.42137608	53.00063291
1016.045454	50.80227269	52.96046072
1064.36826	53.21841302	53.21841302
1113.658217	55.68291083	55.68291083
1165.987833	58.29939164	58.29939164
1221.592285	61.07961424	61.07961424
1280.728722	64.03643612	64.03643612
1343.678687	67.18393436	67.18393436
1410.750833	70.53754164	70.53754164
1482.283997	74.11419986	74.11419986
1558.650678	77.93253389	77.93253389
1640.260966	82.01304829	82.01304829
1727.567015	86.37835073	86.37835073
1821.068118	91.0534059	91.0534059
1921.3165	96.06582501	96.06582501
2028.923925	101.4461962	101.4461962

K1 Gap (TW)	% to K1
14.29823175	0.178262543
14.29644782	0.178365068
14.2941959	0.178494489
14.29153137	0.178647622
14.28848737	0.178822565
14.28508459	0.179018127
14.28133617	0.179233554
14.27725033	0.179468372
14.27283199	0.179722299
14.26808372	0.179995189
14.26300642	0.180286988
14.2575998	0.180597713
14.25186274	0.180927429
14.2457935	0.181276236
14.23938999	0.181644254
14.2326499	0.182031615
14.2255709	0.182438454
14.21815069	0.182864903
14.21038722	0.183311079
14.20227872	0.183777085
14.19382383	0.184262998
14.18502175	0.184768865
14.17587227	0.185294697
14.16637593	0.185840464
14.15653409	0.186406087
14.14634906	0.186991433
14.13582421	0.18759631
14.12496406	0.188220456
14.11377441	0.18886354
14.10226246	0.189525146
14.09043695	0.190204773

14.07830825	0.190901825
14.06588852	0.191615602
14.05319184	0.192345296
14.04023434	0.19308998
14.02703435	0.1938486
14.01361253	0.194619969
13.99999203	0.195402757
13.98609893	0.196201211
13.97134905	0.197048905
13.95567611	0.197949649
13.93900763	0.198907607
13.92126425	0.199927342
13.90235897	0.201013852
13.88219629	0.202172627
13.86067121	0.203409701
13.83766814	0.204731716
13.81305964	0.206145998
13.78670497	0.207660634
13.75844848	0.20928457
13.72811773	0.211027717

GLOBAL STAGNATION COST BY COUNTRY — FULL 200-NATIO

Total planetary stagnation cost: $707.3B/yr at $50/tCO$_2$

Rank	Country	ISO	Stagnation Cost ($M/yr)
1	China	CHN	280113
2	United States	USA	84259
3	India	IND	71813
4	Russia	RUS	26966
5	Japan	JPN	24568
6	South Korea	KOR	12993
7	Iran	IRN	12837
8	Indonesia	IDN	12637
9	Taiwan	TWN	9164
10	South Africa	ZAF	8882
11	Mexico	MEX	8580
12	Turkey	TUR	8400
13	Germany	DEU	8248
14	Australia	AUS	7799
15	Vietnam	VNM	7352
16	Egypt	EGY	6819
17	Malaysia	MYS	5990
18	Canada	CAN	5902
19	Thailand	THA	5534
20	Iraq	IRQ	5180
21	Poland	POL	5098
22	Kazakhstan	KAZ	4752
23	Uzbekistan	UZB	4274
24	Brazil	BRA	3955
25	Bangladesh	BGD	3950
26	Philippines	PHL	3854
27	Italy	ITA	3769
28	Pakistan	PAK	3626
29	United Kingdom	GBR	3178
30	Algeria	DZA	3051
31	Kuwait	KWT	2793
32	Argentina	ARG	2630
33	Spain	ESP	2212
34	Turkmenistan	TKM	2207
35	Israel	ISR	2110
36	Qatar	QAT	1768
37	Netherlands	NLD	1711
38	Bahrain	BHR	1632
39	Czechia	CZE	1516
40	Libya	LBY	1458
41	Ukraine	UKR	1396
42	Serbia	SRB	1292

43	Hong Kong	HKG	1284
44	Morocco	MAR	1265
45	Colombia	COL	1250
46	France	FRA	1188
47	Chile	CHL	1184
48	Oman	OMN	1182
49	Nigeria	NGA	1019
50	Greece	GRC	918
51	Azerbaijan	AZE	886
52	Peru	PER	840
53	Venezuela	VEN	748
54	Belarus	BLR	738
55	Dominican Republic	DOM	734
56	Myanmar	MMR	703
57	Syria	SYR	685
58	Romania	ROU	624
59	Puerto Rico	PRI	622
60	Jordan	JOR	608
61	Laos	LAO	607
62	Tunisia	TUN	584
63	Ghana	GHA	550
64	Belgium	BEL	546
65	Bulgaria	BGR	523
66	Cuba	CUB	488
67	North Korea	PRK	457
68	Austria	AUT	426
69	Ireland	IRL	396
70	Cambodia	KHM	392
71	Ecuador	ECU	346
72	Mongolia	MNG	338
73	Hungary	HUN	328
74	Portugal	PRT	327
75	Trinidad and Tobago	TTO	324
76	Sweden	SWE	302
77	Bolivia	BOL	289
78	New Zealand	NZL	266
79	Brunei	BRN	250
80	Norway	NOR	232
81	Finland	FIN	231
82	Cote d'Ivoire	CIV	219
83	Senegal	SEN	216
84	Tanzania	TZA	204
85	Denmark	DNK	191
86	Congo	COG	184
87	Guatemala	GTM	184
88	Sudan	SDN	180
89	Honduras	HND	170
90	Panama	PAN	166

91	Moldova	MDA	164
92	North Macedonia	MKD	156
93	Angola	AGO	150
94	Cyprus	CYP	140
95	Kyrgyzstan	KGZ	140
96	Slovakia	SVK	138
97	Slovenia	SVN	136
98	Jamaica	JAM	126
99	Mozambique	MOZ	125
100	Zimbabwe	ZWE	124
101	Papua New Guinea	PNG	122
102	Cameroon	CMR	119
103	Croatia	HRV	116
104	Switzerland	CHE	115
105	Botswana	BWA	110
106	Armenia	ARM	108
107	Zambia	ZMB	108
108	Tajikistan	TJK	106
109	Mauritius	MUS	104
110	Georgia	GEO	102
111	Estonia	EST	96
112	Yemen	YEM	90
113	New Caledonia	NCL	88
114	Mali	MLI	86
115	Lebanon	LBN	84
116	Uruguay	URY	84
117	Gabon	GAB	68
118	Bahamas	BHS	67
119	Madagascar	MDG	63
120	Nicaragua	NIC	63
121	Lithuania	LTU	62
122	Guam	GUM	55
123	Paraguay	PRY	55
124	Kenya	KEN	54
125	Malta	MLT	52
126	Burkina Faso	BFA	48
127	Equatorial Guinea	GNQ	48
128	Latvia	LVA	44
129	Guyana	GUY	42
130	Suriname	SUR	41
131	Mauritania	MRT	40
132	Costa Rica	CRI	39
133	Guinea	GIN	37
134	El Salvador	SLV	35
135	Barbados	BRB	33
136	Benin	BEN	30
137	Iceland	ISL	28
138	Aruba	ABW	28

139	Haiti	HTI	27
140	Niger	NER	27
141	Maldives	MDV	26
142	Palestine	PSE	24
143	Cayman Islands	CYM	22
144	Togo	TGO	22
145	Democratic Republic of Cor	COD	22
146	Ethiopia	ETH	22
147	United States Virgin Islands	VIR	22
148	Bermuda	BMU	20
149	Seychelles	SYC	18
150	East Timor	TLS	17
151	Gambia	GMB	17
152	Uganda	UGA	16
153	Fiji	FJI	16
154	Rwanda	RWA	16
155	French Polynesia	PYF	16
156	Bhutan	BTN	14
157	Eritrea	ERI	13
158	Saint Lucia	LCA	13
159	Nepal	NPL	12
160	Cape Verde	CPV	12
161	Chad	TCD	12
162	Albania	ALB	11
163	Antigua and Barbuda	ATG	11
164	Macao	MAC	11
165	Somalia	SOM	11
166	Luxembourg	LUX	10
167	Faroe Islands	FRO	8
168	Liberia	LBR	8
169	Turks and Caicos Islands	TCA	8
170	Grenada	GRD	8
171	Saint Kitts and Nevis	KNA	7
172	Gibraltar	GIB	6
173	Afghanistan	AFG	6
174	American Samoa	ASM	6
175	British Virgin Islands	VGB	6
176	Malawi	MWI	5
177	Burundi	BDI	4
178	Comoros	COM	4
179	Djibouti	DJI	4
180	Dominica	DMA	4
181	Namibia	NAM	4
182	Saint Vincent and the Grena	VCT	4
183	Eswatini	SWZ	4
184	Belize	BLZ	4
185	Solomon Islands	SLB	4
186	Greenland	GRL	3

187	Samoa	WSM	3
188	Guinea-Bissau	GNB	2
189	Sao Tome and Principe	STP	2
190	Tonga	TON	2
191	Vanuatu	VUT	2
192	Nauru	NRU	2
193	Saint Pierre and Miquelon	SPM	2
194	Kiribati	KIR	1
195	Cook Islands	COK	0
196	Falkland Islands	FLK	0
197	Lesotho	LSO	0
198	Montserrat	MSR	0
199	Saint Helena	SHN	0
200	Sierra Leone	SLE	0
	GLOBAL TOTAL		**707335**

N TABLE

% of Global	Cumulative %	% of National GDP	Elec Gen (TWh)
39.6	39.6	1.039	10086.9
11.91	51.5	0.432	4391
10.15	61.7	0.685	2030.2
3.81	65.5	0.723	1209.3
3.47	69	0.515	1016.4
1.84	70.8	0.608	625.4
1.81	72.6	0.817	395.8
1.79	74.4	0.361	371.5
1.3	75.7	0.734	288.6
1.26	76.9	1.227	249.1
1.21	78.2	0.422	355.2
1.19	79.3	0.352	353.9
1.17	80.5	0.211	497.4
1.1	81.6	0.58	281.6
1.04	82.6	0.882	303.6
0.96	83.6	0.467	237.4
0.85	84.5	0.683	198.2
0.83	85.3	0.335	636.8
0.78	86.1	0.492	199.5
0.73	86.8	0.982	150.3
0.72	87.5	0.414	172.2
0.67	88.2	0.944	118.5
0.6	88.8	1.086	76.2
0.56	89.4	0.124	745.7
0.56	89.9	0.46	113.9
0.54	90.5	0.38	125.9
0.53	91	0.176	264.2
0.51	91.5	0.275	182.1
0.45	92	0.122	292.8
0.43	92.4	0.512	96.3
0.39	92.8	1.276	87.7
0.37	93.2	0.308	152.5
0.31	93.5	0.136	288.6
0.31	93.8	2.264	33.8
0.3	94.1	0.627	74.4
0.25	94.3	0.463	58.6
0.24	94.6	0.195	135.1
0.23	94.8	2.777	36.2
0.21	95	0.442	75.5
0.21	95.2	1.594	35.1
0.2	95.4	0.452	111.5
0.18	95.6	1.106	37.1

0.18	95.8	0.36	37.6
0.18	96	0.406	43.9
0.18	96.1	0.168	87.5
0.17	96.3	0.044	568.3
0.17	96.5	0.264	89.3
0.17	96.6	0.666	43.3
0.14	96.8	0.09	40.1
0.13	96.9	0.348	58.2
0.13	97	0.506	28
0.12	97.2	0.246	63.8
0.11	97.3	0.499	82.9
0.1	97.4	0.42	45.7
0.1	97.5	0.384	25.9
0.1	97.6	0.242	24.7
0.1	97.7	1.308	20.1
0.09	97.8	0.124	49.8
0.09	97.9	0.546	18.7
0.09	97.9	0.613	22.6
0.09	98	1.105	52.3
0.08	98.1	0.451	20.8
0.08	98.2	0.378	24.3
0.08	98.3	0.112	72.9
0.07	98.3	0.378	38
0.07	98.4	0.58	15.3
0.06	98.5	1.129	26.5
0.06	98.5	0.107	74.7
0.06	98.6	0.129	31
0.06	98.6	0.601	15.7
0.05	98.7	0.198	33
0.05	98.7	0.755	8.4
0.05	98.8	0.115	40.5
0.05	98.8	0.111	51.2
0.05	98.9	0.864	9.5
0.04	98.9	0.061	170.7
0.04	99	0.37	12.4
0.04	99	0.133	44.5
0.04	99		5.6
0.03	99.1	0.048	159.8
0.03	99.1	0.102	81.3
0.03	99.1	0.18	11.1
0.03	99.2	0.415	8.1
0.03	99.2	0.103	11
0.03	99.2	0.064	33.4
0.03	99.2	0.779	5.2
0.03	99.3	0.125	13.5
0.03	99.3		16.8
0.02	99.3	0.341	11.7
0.02	99.3	0.176	12.9

0.02	99.4	0.688	5.2
0.02	99.4	0.544	5.9
0.02	99.4	0.095	17.9
0.02	99.4	0.409	5.7
0.02	99.4	0.442	16.2
0.02	99.5	0.092	29.1
0.02	99.5	0.197	14.9
0.02	99.5	0.629	4.5
0.02	99.5	0.398	19.6
0.02	99.5	0.479	8.3
0.02	99.6		4.7
0.02	99.6	0.144	8.3
0.02	99.6	0.114	14.5
0.02	99.6	0.021	61.7
0.02	99.6	0.28	2.6
0.02	99.6	0.272	8.9
0.02	99.6	0.158	19.5
0.01	99.7	0.21	21.7
0.01	99.7	0.383	3.3
0.01	99.7	0.145	14.3
0.01	99.7	0.241	6.1
0.01	99.7	0.148	3.1
0.01	99.7		3.1
0.01	99.7	0.281	4.4
0.01	99.8	0.171	4.5
0.01	99.8	0.121	17.3
0.01	99.8	0.19	3.2
0.01	99.8		2
0.01	99.8	0.163	2.6
0.01	99.8	0.192	4.4
0.01	99.8	0.073	8.8
0.01	99.8		1.8
0.01	99.8	0.084	44.2
0.01	99.8	0.029	12.9
0.01	99.8	0.281	2.2
0.01	99.9	0.129	1.7
0.01	99.9	0.235	1.6
0.01	99.9	0.085	6.3
0.01	99.9		1.3
0.01	99.9		2.1
0.01	99.9	0.252	1.6
0.01	99.9	0.049	12.4
0.01	99.9	0.156	4.1
0	99.9	0.059	7.1
0	99.9	0.968	1.1
0	99.9	0.089	1
0	99.9	0.178	20.1
0	99.9		1

0	99.9	0.154	1
0	99.9	0.109	0.8
0	99.9		0.9
0	99.9	0.098	1
0	99.9		0.7
0	99.9	0.154	0.9
0	99.9	0.024	15.9
0	99.9	0.008	18.3
0	99.9		0.7
0	99.9		0.6
0	99.9	0.59	0.6
0	100		0.5
0	100	0.365	0.5
0	100	0.016	5.8
0	100		1.1
0	100	0.056	1.1
0	100		0.7
0	100		11.2
0	100		0.4
0	100	0.743	0.4
0	100	0.013	10.7
0	100	0.282	0.5
0	100	0.05	0.4
0	100	0.03	9
0	100		0.4
0	100		0.5
0	100		0.4
0	100	0.026	1.5
0	100		0.5
0	100	0.201	0.4
0	100		0.3
0	100		0.2
0	100		0.2
0	100		0.2
0	100	0.011	1
0	100		0.2
0	100		0.2
0	100	0.019	1.8
0	100	0.049	0.4
0	100	0.268	0.1
0	100	0.133	0.2
0	100	0.667	0.2
0	100	0.021	1.9
0	100		0.2
0	100	0.042	0.6
0	100		0.4
0	100		0.1
0	100		0.5

0	100		0.2
0	100	0.077	0.1
0	100	0.303	0.1
0	100		0.1
0	100		0.1
0	100		0
0	100		0.1
0	100		0
0	100		0
0	100		0
0	100	0.011	0.5
0	100		0
0	100		0
0	100	0.004	0.2
100			

Carbon Intensity (gCO_2/kWh)	Fossil Share %	K1 ETA (yrs)	Gov Efficiency
555	62	21	0.2376
384	58	35	0.2012
707	78	42	0.2156
446	64	32	0.2393
483	69	6	0.2993
416	60	33	0.2257
649	93	18	0.2093
680	82	24	0.1728
635	85	30	0.2702
713	84	83	0.1549
483	75	20	0.2264
475	57	274	0.1139
332	41	51	0.3137
554	65	1750	0.0327
484	58	87	0.1363
574	89	335	0.0849
604	80	617	0.1158
185	22	27	0.275
555	85	230	0.1384
689	99	274	0.1392
592	69	323	0.1397
802	85	19	0.2013
1121	91	91	0.1435
106	11	5	0.2603
694	98	128	0.1352
612	78	144	0.1304
285	51	239	0.1101
398	54	14	0.2494
217	36	25	0.1983
634	99	51	0.1736
637	98	184	0.1388
345	59	11	0.2544
153	25	15	0.1918
1306	100	229	0.0342
567	89	26	0.126
603	100	1047	0.0781
253	46	32	0.2408
902	100	93	0.1212
401	41	37	0.2354
831	100	1442	0.1111
250	28	12	0.2021
696	73	73	0.1694

682	99	214	0.164
577	73	44	0.1554
286	36	44	0.1385
42	5	18	0.2609
265	30	274	0.1111
545	96	161	0.1788
508	77	1679	0.1529
316	50	369	0.1366
633	88	36	0.1814
263	41	343	0.1606
180	22	130	0.1512
323	63	6	0.2122
566	81	350	0.1464
570	60	193	0.1544
682	96	128	0.1625
251	32	31	0.2053
665	94	125	0.18
539	77	50	0.1434
232	23	25	0.1363
560	96	259	0.1608
453	61	78	0.1298
150	28	110	0.1902
275	28	18	0.23
639	95	896	0.1213
344	37	163	0.1458
114	16	107	0.1175
256	52	35	0.1378
498	59	162	0.1294
210	28	212	0.1602
808	90	528	0.1542
162	25	30	0.1887
128	19	111	0.0851
682	100	101	0.1593
35	1	40	0.1631
468	62	75	0.1553
120	15	324	0.1504
893	100	123	0.1561
29	1	54	0.1666
57	4	6	0.1932
394	69	84	0.1742
535	78	19	0.181
372	74	195	0.1357
114	9	190	0.1406
714	79	49	0.1322
273	25	20	0.1623
214	30	648	0.1278
290	38	29	0.1421
259	38	21	0.1674

630	88	22	0.1631
531	63	347	0.1633
167	24	35	0.1502
488	72	352	0.0989
173	17	287	0.1337
95	15	133	0.2211
183	21	62	0.2384
561	87	48	0.1583
128	16	36	0.175
298	32	47	0.1564
514	76	128	0.1627
286	36	143	0.1374
160	24	109	0.1155
37	3	448	0.1291
849	100	16	0.1756
243	40	399	0.1174
111	11	15	0.1771
98	9	92	0.2596
633	83	60	0.1739
143	20	42	0.1582
319	40	40	0.1401
586	83	250	0.0813
566	74	152	0.1568
394	57	444	0.1727
369	53	73	0.206
97	5	18	0.1227
429	52	1029	0.0806
654	99	278	0.0371
477	65	172	0.1411
288	35	137	0.1584
140	23	358	0.1374
611	92	134	0.0676
25		34	0.0826
85	8	469	0.1271
472	82	52	0.1839
555	83	78	0.1699
605	69	879	0.154
139	27	22	0.155
634	93	34	0.1868
383	57	73	0.1528
482	73	60	0.1817
63	6	28	0.1571
183	25	66	0.1235
99	7	112	0.1597
600	92	193	0.1739
590	97	75	0.0888
28		20	0.1089
550	83	110	0.0861

535	81	278	0.1946
675	97	40	0.1567
612	93	96	0.0698
461	67	722	0.0705
643	97	245	0.0465
478	79	72	0.1406
27		11	0.188
24		19	0.1343
642	97	102	0.0644
641	98	4465	0.2747
571	86	19	0.1172
667	100		0
667	100		0
57	3	43	0.1424
278	37	42	0.158
302	43	73	0.1399
437	66	1986	0.1333
24			0
591	89	80	0.0557
650	97	144	0.0397
23		24	0.2052
480	72	599	0.1983
615	95	46	0.071
24		195	0.159
611	94	30	0.0806
449	59	26	0.1577
524	81	170	0.0542
124	8	299	0.1656
354	54	64	0.155
436	67	179	0.0491
654	100		0
667	100		0
636	95	68	0.0751
591	100	75	0.0652
124	13	282	0.1668
647	100	99	0.0593
647	100		0
55	4	38	0.1592
231	31	82	0.1009
643	100		0
450	65	646	0.0255
600	87	552	0.1788
48	2	8	0.1681
600	87	1189	0.2023
143	4	32	0.1631
156	11	33	0.1625
636	91	752	0.023
111	13	373	0.1649

400	60	47	0.1127
625	100		0
556	89	38	0.1632
571	86	241	0.0359
500	75	20	0.0961
750	100		0
600	100		0
500	75	681	0.0249
250	50	326	0.0454
1000	100	217	0.0674
21			0
1000	100		0
1000	100		0
48	5	85	0.1639

1% Redirect ($M)	5% Redirect ($M)	Development Stage
2801.1	14005.6	Early transition
842.6	4213	Mid-transition
718.1	3590.7	Early transition
269.7	1348.3	Early transition
245.7	1228.4	Early transition
129.9	649.6	Early transition
128.4	641.9	Degrading (moving away from K1)
126.4	631.9	Early transition
91.6	458.2	Early transition
88.8	444.1	Early transition
85.8	429	Early transition
84	420	Early transition
82.5	412.4	Mid-transition
78	390	Early transition
73.5	367.6	Early transition
68.2	341	Pre-transition (fossil locked)
59.9	299.5	Early transition
59	295.1	Active transition
55.3	276.7	Early transition
51.8	259	Early transition
51	254.9	Early transition
47.5	237.6	Early transition
42.7	213.7	Early transition
39.5	197.7	Post-fossil (K1 approaching)
39.5	197.5	Early transition
38.5	192.7	Early transition
37.7	188.4	Early transition
36.3	181.3	Mid-transition
31.8	158.9	Active transition
30.5	152.6	Early transition
27.9	139.7	Pre-transition (fossil locked)
26.3	131.5	Mid-transition
22.1	110.6	Mid-transition
22.1	110.4	Pre-transition (fossil locked)
21.1	105.5	Early transition
17.7	88.4	Pre-transition (fossil locked)
17.1	85.6	Mid-transition
16.3	81.6	Early transition
15.2	75.8	Mid-transition
14.6	72.9	Pre-transition (fossil locked)
14	69.8	Mid-transition
12.9	64.6	Early transition

12.8	64.2	Early transition
12.7	63.2	Early transition
12.5	62.5	Early transition
11.9	59.4	Post-fossil (K1 approaching)
11.8	59.2	Early transition
11.8	59.1	Early transition
10.2	51	Early transition
9.2	45.9	Early transition
8.9	44.3	Early transition
8.4	42	Mid-transition
7.5	37.4	Early transition
7.4	36.9	Early transition
7.3	36.7	Early transition
7	35.1	Early transition
6.9	34.2	Early transition
6.2	31.2	Mid-transition
6.2	31.1	Early transition
6.1	30.4	Early transition
6.1	30.4	Early transition
5.8	29.2	Early transition
5.5	27.5	Early transition
5.5	27.3	Active transition
5.2	26.2	Active transition
4.9	24.4	Early transition
4.6	22.9	Early transition
4.3	21.3	Post-fossil (K1 approaching)
4	19.8	Early transition
3.9	19.6	Early transition
3.5	17.3	Early transition
3.4	16.9	Early transition
3.3	16.4	Active transition
3.3	16.4	Post-fossil (K1 approaching)
3.2	16.2	Early transition
3	15.1	Post-fossil (K1 approaching)
2.9	14.5	Early transition
2.7	13.3	Post-fossil (K1 approaching)
2.5	12.5	Early transition
2.3	11.6	Post-fossil (K1 approaching)
2.3	11.6	Post-fossil (K1 approaching)
2.2	11	Early transition
2.2	10.8	Early transition
2	10.2	Degrading (moving away from K1)
1.9	9.6	Post-fossil (K1 approaching)
1.8	9.2	Degrading (moving away from K1)
1.8	9.2	Early transition
1.8	9	Early transition
1.7	8.5	Early transition
1.7	8.3	Mid-transition

1.6	8.2	Early transition
1.6	7.8	Early transition
1.5	7.5	Early transition
1.4	7	Early transition
1.4	7	Post-fossil (K1 approaching)
1.4	6.9	Post-fossil (K1 approaching)
1.4	6.8	Mid-transition
1.3	6.3	Early transition
1.2	6.2	Post-fossil (K1 approaching)
1.2	6.2	Early transition
1.2	6.1	Early transition
1.2	6	Early transition
1.2	5.8	Early transition
1.2	5.8	Post-fossil (K1 approaching)
1.1	5.5	Early transition
1.1	5.4	Early transition
1.1	5.4	Post-fossil (K1 approaching)
1.1	5.3	Post-fossil (K1 approaching)
1	5.2	Degrading (moving away from K1)
1	5.1	Post-fossil (K1 approaching)
1	4.8	Early transition
0.9	4.5	Pre-transition (fossil locked)
0.9	4.4	Early transition
0.9	4.3	Early transition
0.8	4.2	Early transition
0.8	4.2	Post-fossil (K1 approaching)
0.7	3.4	Early transition
0.7	3.4	Pre-transition (fossil locked)
0.6	3.2	Degrading (moving away from K1)
0.6	3.2	Early transition
0.6	3.1	Early transition
0.6	2.8	Pre-transition (fossil locked)
0.6	2.8	Post-fossil (K1 approaching)
0.5	2.7	Post-fossil (K1 approaching)
0.5	2.6	Early transition
0.5	2.4	Early transition
0.5	2.4	Early transition
0.4	2.2	Early transition
0.4	2.1	Early transition
0.4	2.1	Early transition
0.4	2	Early transition
0.4	2	Post-fossil (K1 approaching)
0.4	1.9	Early transition
0.4	1.8	Post-fossil (K1 approaching)
0.3	1.7	Early transition
0.3	1.5	Pre-transition (fossil locked)
0.3	1.4	Post-fossil (K1 approaching)
0.3	1.4	Pre-transition (fossil locked)

0.3	1.4	Early transition
0.3	1.4	Early transition
0.3	1.3	Pre-transition (fossil locked)
0.2	1.2	Early transition
0.2	1.1	Pre-transition (fossil locked)
0.2	1.1	Degrading (moving away from K1)
0.2	1.1	Post-fossil (K1 approaching)
0.2	1.1	Post-fossil (K1 approaching)
0.2	1.1	Pre-transition (fossil locked)
0.2	1	Early transition
0.2	0.9	Early transition
0.2	0.9	Pre-transition (fossil locked)
0.2	0.9	Pre-transition (fossil locked)
0.2	0.8	Post-fossil (K1 approaching)
0.2	0.8	Early transition
0.2	0.8	Early transition
0.2	0.8	Early transition
0.1	0.7	Post-fossil (K1 approaching)
0.1	0.7	Pre-transition (fossil locked)
0.1	0.7	Pre-transition (fossil locked)
0.1	0.6	Post-fossil (K1 approaching)
0.1	0.6	Early transition
0.1	0.6	Pre-transition (fossil locked)
0.1	0.6	Post-fossil (K1 approaching)
0.1	0.6	Unstable (burst-driven, no clear path)
0.1	0.6	Early transition
0.1	0.6	Pre-transition (fossil locked)
0.1	0.5	Post-fossil (K1 approaching)
0.1	0.4	Early transition
0.1	0.4	Unstable (burst-driven, no clear path)
0.1	0.4	Pre-transition (fossil locked)
0.1	0.4	Pre-transition (fossil locked)
0.1	0.4	Unstable (burst-driven, no clear path)
0.1	0.3	Unstable (burst-driven, no clear path)
0.1	0.3	Post-fossil (K1 approaching)
0.1	0.3	Unstable (burst-driven, no clear path)
0.1	0.3	Pre-transition (fossil locked)
0.1	0.2	Post-fossil (K1 approaching)
0	0.2	Early transition
0	0.2	Pre-transition (fossil locked)
0	0.2	Unstable (burst-driven, no clear path)
0	0.2	Degrading (moving away from K1)
0	0.2	Post-fossil (K1 approaching)
0	0.2	Early transition
0	0.2	Post-fossil (K1 approaching)
0	0.2	Post-fossil (K1 approaching)
0	0.2	Pre-transition (fossil locked)
0	0.2	Post-fossil (K1 approaching)

0	0.2	Early transition
0	0.1	Pre-transition (fossil locked)
0	0.1	Early transition
0	0.1	Pre-transition (fossil locked)
0	0.1	Unstable (burst-driven, no clear path)
0	0.1	Pre-transition (fossil locked)
0	0.1	Pre-transition (fossil locked)
0	0.1	Unstable (burst-driven, no clear path)
0	0	Unstable (burst-driven, no clear path)
0	0	Unstable (burst-driven, no clear path)
0	0	Post-fossil (K1 approaching)
0	0	Pre-transition (fossil locked)
0	0	Pre-transition (fossil locked)
0	0	Post-fossil (K1 approaching)
7073	**35367**	